Household credit and women's outcomes: Experimental evidence from India*

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Abstract

Women in rural agrarian households are constrained in their ownership of individual assets and their access to financial resources. We test the effects of relaxing households' liquidity constraints through the randomized provision of collateral-free loans to 1,429 farming households across 80 villages. Since Indian farming households rely on joint contributions from men and women, and the loan increased household ownership of goods that benefit both members. We explore its impacts on household members' individual residual income and private consumption. The loan product enhanced women's financial autonomy and uptake of digital financial services, accompanied by an expansion of women's access to resources. Women in households that received the loan allocated more hours to leisure and fewer to farm and non-farm work on the busiest day of the agricultural season than the control group, although women's labor supply across the agricultural season and household farm income did not change between treatment and control villages. The loan did not affect women's intrahousehold bargaining power but amplified pre-existing cooperative household arrangements. The results suggest that credit to households with joint production can improve women's outcomes, such as access to resources and financial inclusion, that are often difficult to affect through targeted resource provision.

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1 Introduction

Who in the household receives resources is a central question for models of household behavior and for the design of social assistance. Many of the largest cash transfer and microfinance programs in low- and middle-income countries designate women as recipients, on the premise that their spending priorities align with program goals. Yet these programs don't always improve their economic outcomes or agency within the household (Banerjee et al., 2015; Roy et al., 2015; Green et al., 2015; Attanasio et al., 2015; Angelucci et al., 2015; Said et al., 2019). One explanation is that directing resources to women threatens the existing distribution of power, prompting resistance from other household members. Loan funds are appropriated (Fiala, 2018; Bernhardt et al., 2019) and cash transfers provoke male backlash (Angelucci, 2008; Bobonis et al., 2013), underscoring the risks that targeted interventions pose to women. By contrast, a within-household transfer of assets to women strengthened their control over resources and involvement in decisions without disrupting household production (Ambler et al., 2026). An important question to ask here is can programs that reach women without targeting them be successful? This matters most in resource-poor settings, where marginal returns from social protection and credit are largest, but also, the pecuniary and social costs of targeting are hardest to bear.

We evaluate the impact of a widely used policy instrument, collateral-free agricultural credit, on women's outcomes in India. Agricultural production in India is organized jointly at the household level, with a division of labor across tasks: most farm managers are men, while women typically undertake labor-intensive activities such as weeding, harvesting, and livestock management (Afridi et al., 2023; Gulati et al., 2025). Targeting an agricultural loan to women may therefore not change how households use the funds. The program increased asset accumulation but did not change input use or farm income in our sample[1], so we examine two channels through which it may affect household members, particularly women. In the consumption channel, household spending on composite goods, such as productive assets and consumer durables that members consume both jointly and individually, raises their utility, assuming they value such goods. In the residual income channel, the loan can change the contribution each member makes to household expenditure, and therefore their residual income, with corresponding effects on their financial autonomy and leisure.

In this paper, we utilize data from a cluster randomized controlled trial with 1,429 farming households in 80 villages of India. Households were offered a collateral-free *loan*

[1] Using the full study sample which is an imbalanced panel, Kramer et al. (2025a) find that the loan increased cultivated area, revenue, and profit in Odisha, with limited effects in Maharashtra. Our estimates differ in sample, outcome definition, and specification.

product between 2023 and 2024 in 40 of the study villages (treatment group) as part of the experiment, while the remaining 40 villages (control group) did not receive this product but could access formal and informal credit suppliers in the market. The randomization allows us to measure the causal effect of the loan product on household consumption and individual-level outcomes. The loan amount was roughly less than half of the annual household income and 99% of the beneficiaries repaid it. The loan was disbursed at the start of the planting season, when households' credit needs peak, and had a 14% interest rate, on average 10 percentage points below other credit available in the market. Despite the low interest rate, high repayment rates among borrowers yielded the lending institution an average profit of USD 18 per farmer.

We employ two empirical models in our analysis. First, we estimate the intent-to-treat (ITT) effect of the loan program on household spending using an analysis of covariance (ANCOVA) model. Here we exploit the cluster randomization of the loan program at the village level and a household panel dataset combining baseline and endline surveys. Second, we examine how the loan affected the outcomes of household members, focusing on their financial inclusion, time use, and involvement in decisions. The endline survey interviewed two members per household compared to one at baseline, the same farmer that was interviewed in the baseline survey and one additional household member, yielding a sample of 1,126 women. We use an Instrumental Variables (IV) model in which randomized village-level treatment assignment instruments for households that took up the loan product, henceforth known as "borrower households". Restricting the endline sample to female respondents and controlling for their position in the household, we estimate the Local Average Treatment Effect (LATE) of the program on women's outcomes. We separately estimate the impact on men and report it alongside women's results.

The program increased household accumulation of productive capital and consumer durables. The treatment group was 14 percentage points (pp) more likely to own non-mechanized farm equipment and 8 pp more likely to own large consumer durables than the control group. Diet diversity also improved as the treatment group's food consumption score was 17% greater than the control group's (39.4 vs. 33.8). Building on the household's spending response to credit, we use individual-level data and a model of intrahousehold allocation under joint production to trace how the loan product affected household members, specifically women.

The program resulted in two key impacts on women. First, it positively impacted financial inclusion and autonomy. Members in borrower households transacted from their bank account more regularly, and women were 40 pp more likely to receive deposits

other than earnings or government transfers, suggesting changes in their residual income. Accompanied by expanded resource access, women exercised more financial autonomy. They were 70 pp more likely in borrower households to use the debit cards linked with their individual bank account by themselves than the control group. They were about 81 pp more likely to use mobile apps for banking. Both independent use of debit cards and mobile banking indicate financial autonomy through greater privacy of transactions. Men also experienced gains in financial autonomy. The effects for men are positive but smaller than those for women, since men's control group means are higher.

Second, women responded to the loan product by allocating more time to leisure activities (2.6 versus 1.7 hours in the control group) on the busiest day of the agricultural season. They reported fewer hours than the control group in farm work during this peak time, however, their contribution in the number of days over the entire season was not significantly different. Real wages paid for hiring farm labor for that agricultural season and annual real income from farming did not change suggesting that households' ownership of non-mechanized farm equipment may have increased women's productivity. Women reduced their labor supply to business or wage employment. Our data does not let us distinguish the mechanisms behind this reduction. There are two possible explanations. One, it could be the response of households no longer requiring additional income to supplement their budget because of the loan. Two, the decline was concentrated among households in the state of Maharashtra, and systematic social norms around women's non-agricultural labor supply could also possibly explain this pattern. In households with higher ex-ante coordination among spouses, the loan's effect on women's supply to non-farm work was significantly larger than the rest, suggesting that the loan supported women's non-farm activities in cooperative households. We find consistent patterns in women's recall of time allocations in the day prior to the survey. Men's time allocation to production or leisure did not differ between the loan product-receiving households and the control group.

The loan did not shift the balance of power between spouses as measured by women's or men's involvement in household production and consumption decisions. However, it reinforced existing cooperative household arrangements. Greater spousal coordination among households prior to the intervention corresponded positively with women's involvement in decisions on non-agricultural production activities, domestic work, household purchases, their private consumption and personal activities.

This paper contributes to the literature on women's financial inclusion and autonomous control over resources, highlighting that reducing households' liquidity constraints is an

important channel to improve these outcomes. This paper finds a large impact on women's knowledge and use of mobile banking, which is striking given the otherwise low uptake of digital finance in rural India because of rigid gender norms (Barboni et al., 2018). Our results stand in contrast to those of a targeted smartphone distribution to rural women in India, in which 40 percent of women transferred the device to another household member and the program had no aggregate effect on their knowledge or use of mobile banking (Barboni et al., 2024). Together, these findings highlight the value of program designs that improve women's financial outcomes without explicit targeting.

This paper contributes to the literature evaluating credit programs and their effects on women. Our intervention differed from the microfinance programs usually studied in this context as it did not involve group-based saving and lending, included no bundled training that could influence women's subjective well-being or empowerment, and did not target borrowers by gender. Microcredit programs for women that operate through group-based lending have had mixed effects on women's decision-making ability, leaving their impact on women's empowerment inconclusive. There were no improvements in decisions over household spending, investment, savings, or children's health and education in India (Banerjee et al., 2015), and none on a female empowerment index that captured women's capacity to make consumption and production decisions and her mobility in Morocco (Crépon et al., 2015). By contrast, there was an increase in women's involvement in decisions in Mexico (Angelucci et al., 2015). We find no aggregate impacts on women's involvement in decision making but greater engagement in households where women were already involved ind decision making.

Outside of group-based saving and lending, credit programs have often failed to improve women's economic outcomes. For instance, loans to women in multi-adult households raised household income and other members' enterprise profits, but not women's own profits (de Mel et al., 2008; Garikipati, 2008; Fiala, 2018; Friedson-Ridenour and Pierotti, 2019; Bernhardt et al., 2019). Other financial interventions that are designed to be less susceptible to resource capture are more successful, further motivating this paper's investigation. For instance, savings-led programs improved women's empowerment indices in rural Ghana, Malawi and Uganda (Karlan et al., 2017) and in the Philippines (Ashraf et al., 2010).

This paper also adds to the literature on how social protection and credit programs affect women's labor supply. In our setting, the loan increased women's time spent on leisure and reduced their time in non-farm work, plausibly from more productive supply of labor, less requirement to supplement household budget and/ or social norms. Credit has increased labor supply (Fink et al., 2020) while transfers have reduced women's prob-

ability of working and increased men's participation in domestic chores (Hidrobo et al., 2016; Beaman et al., 2023; El-Enbaby et al., 2025). Programs directly impacting agricultural productivity, such as the System of Rice Intensification, caused women to reallocate labor from non-farm activities into rice cultivation in Indonesia (Takahashi and Barrett, 2014).

Governments in developing countries routinely provide agricultural loan subsidies or collateral-free credit to encourage investment among small and marginal farmers. The effects on household welfare, and on individual members in particular, remain understudied, a gap made more important by the mixed evidence on agricultural outcomes (Karlan et al., 2014; Tarozzi et al., 2015; Maitra et al., 2017; Nakano and Magezi, 2020; Fink et al., 2020; Beaman et al., 2023; Mukherjee et al., 2024). This paper builds on Kramer et al. (2025a) which reports how the credit program impacted indebtedness, renewal and farm profitability among other outcomes. Our findings on how the collateral-free loan program affected household members generates an important policy implication: household-level interventions in joint production structures can be utilized as an effective way to improve women's outcomes while avoiding the pecuniary and social costs of targeted programs. This is critical for development finance institutions and governments that are actively trying to scale up such products in low-and-middle income countries.

2 Theoretical framework: Loan and Individual Welfare

The agricultural sector in India is well-suited to test the indirect effects of a household-level intervention on its members because production involves joint contributions by household members on the same plot. The Indian Ministry of Agriculture & Farmers Welfare formally recognizes this joint production system, classifying agricultural land managed by multiple members of the same household as "individually operated". Doss (2018) documents that it is difficult to separate individual contribution to and profits from agricultural production. Targeting agricultural credit to women in multi-adult households, therefore, may not guarantee an improvement in women's control over resources. In this section, we outline how household members are potentially affected from a program that targets the household's liquidity constraints.

One pathway for the household credit to affect its members through intensification of household production that could affect the household's demand for labor. We describe stylized facts below that rule out agricultural intensification as the main channel, allowing us to focus on the effects of reducing household liquidity through the consumption channel. We also expect that the loan product increased household members' engagement

with digital financial services as it allowed digital repayments.

Stylized Fact 1. *The loan product increased household consumption of composite goods.*

Bobonis (2009) characterizes a household's consumption as public, private and "composite" where composite goods are consumed both individually and jointly by household members. We find increased accumulation of composite goods in our setting through farm and business equipment and consumer durables. We report estimates of a formal test in section 5.1.

Stylized Fact 2. *The loan product did not change a household's spending on wages or other inputs.*

In our sample, we find no detectable changes between treatment and control villages in the wage rate paid for hired labor, spending on other inputs, or real farm income.[2] There were also no differences in the total number of days worked by family members over the season. We therefore rule out the agricultural intensification channel in theorizing impacts of the loan on household members.

We now theorize impacts on household members taking these stylized facts into account. We utilize the collective household model assuming that household members have distinct preferences following Chiappori (1988). This lets us identify the loan's effect on individual household members through both household-level and private consumption channels, and test the impacts on women and men separately.[3]

Appendix A describes the full set up and propositions of our static model of household and individual decision making in response to the loan. We consider a two member household where each member has a distinct individual preferences and earns a labor and non-labor income. Each individual derives a positive utility from household consumption as well as their own private consumption. The total household budget is a sum of labor and non-labor income as well as joint non-labor income. The following assumptions are relevant to our research questions and testable hypotheses.

Assumption 1. *The household treats the loan as an increase in joint, non-labor income in the short term.*

[2] Estimates are reported in Appendix C and discussed alongside results in section 5.2. The companion paper (Kramer et al., 2025a) evaluates the impact of the loan program on value of production and returns on investment. In this paper, we focus on estimating household's spending on input utilization to interpret the effects we find on household members' labor supply.

[3] A unitary model assumes that a household's use of the loan would not depend on which member is the beneficiary of the product (Samuelson, 1956), the household has a single decision maker and household members have homogeneous preferences. Such a framework would not allow us to measure the indirect utility response for household members where household consumption increases and but their residual income reduces.

Assumption 1 reflects the organization of farm production at the household level, which makes a loan for farming likely to be treated as a joint resource. The program design is consistent with this, since two household members co-signed the loan (Section 3.3). The static model captures household decisions in the short term when the loan expands the household budget. Although most loans were repaid before the endline survey, the composite goods acquired by the household, such as farm equipment and consumer durables, remained with them and so endline outcomes reflect this expansion (Stylized Fact 1).

Assumption 2. *Financial autonomy and leisure are normal goods for each household member.*

We define financial autonomy as a member's independent access to and use of financial services. If the loan raises a member's residual income, normality implies that they exercise more financial autonomy and consume more leisure. Stylized Fact 2 shows that the loan did not change the household's use of farm inputs, including family labor days, or farm income. Changes in members' time allocation are therefore unlikely to reflect changes in demand for labor in farm production. This inference does not extend to labor demand for non-farm activities.

Given the loan did not change input use, our assumption that leisure is a normal good predicts that if the loan raises individual residual income, hours allocated to leisure would increase and agricultural production would fall.

We set up the household optimization as a two-stage problem in Appendix A that distinguishes the loan's effects on household-level and individual-level consumption. In the first stage, given the household budget, which includes the loan amount, and bargaining weight of household members, the household chooses expenditure on the public and composite goods. In the second stage, conditional on allocations to public and composite goods, each member chooses private consumption to maximize utility subject to an individual residual income constraint. The individual residual income is the difference between the total earnings and contributions to household consumption. We assume that every individual's contribution to the household consumption is a function of their bargaining power. We show that the indirect utility of an individual in response to the loan can be decomposed into the change in utility from increased consumption of public goods, composite goods and changes in their residual income. The loan can change the individual residual income in the short term in two ways. One is a bargaining channel, where if the loan shifts the bargaining rule in member's favor, bargaining power rises and their required contribution falls. The other is the resource expansion channel where as the total household resources grow because of the loan product, the total individual contributions required to finance joint expenditure fall, relaxing both members' private

budget constraints.

2.1 Testable predictions

We analyze first-stage changes in household consumption of composite goods through the ownership of farm and business equipment and consumer durables, and report the household's food consumption score as an additional household-level outcome. In the second stage, we do not directly observe individual residual income, but interpret its effects through members' financial autonomy measured as the independent use of debit cards and mobile apps for banking, and through time allocated to personal activities, farm work, and non-farm work. We test for the resource expansion channel affecting individual residual income through self-reported account activity such as deposits, and the bargaining power channel through their self-reported involvement in decisions on production and consumption.[4]

3 Experimental Design and Data Collection

3.1 Context

This intervention was conducted in two districts of India: Jajpur in the state of Odisha and Amravati in Maharashtra (Figure 1). Farmers in Jajpur mainly grow rice, while those in Amravati grow cotton and soybean. Both groups face credit constraints, based on reports from our implementation partner Dvara E-Registry (DER), a fintech company. The loan program targeted small and marginal farmers[5], which constitute the majority of agricultural holdings in both states.[6] The rice crop in Odisha is transplanted June through July and harvested in September through November. The agriculture sector in this state employs more than 70% of rural women, most of whom are engaged in rice farming (Barooah et al., 2025). Women almost exclusively perform nursery preparation, transplanting, weeding, harvesting, and threshing. Weeding is done manually or with simple hand tools. Cotton is sown in Maharashtra from May to June and harvested between October through December. Women in Maharashtra are responsible for the majority of on-field tasks of cotton cultivation such as sowing seeds, stalk removal, ensuring optimum plant population, weeding, fertilizer application, and picking (Dwivedi and Damle, 2019). Men are primarily responsible for operating machinery (tractors, ox-carts) and for managing market and sales transactions. Soybean is sown June- July and harvested September

[4] We also asked respondents their income shares from farm and non-farm work to test whether women's involvement in decisions on their private consumption is influenced by the share of their income from non-farm work,, however, this question was not administered correctly by the enumerators.

[5] Small farms are plots of size 1-2 hectare and marginal farms have plot sizes less than 1 hectare.

[6] The Agricultural Census 2015-16 classifies over 30% of landholdings in Odisha and Maharashtra as small. Marginal holdings account for 45% of landholdings in Odisha and 18% in Maharashtra.

through October. For soybean, women perform the more labor-intensive manual tasks including weeding, loosening soil, and seed preservation, while men handle ploughing, marketing, and machine operations.

The key constraints to credit access in our study locations include expensive interest rates and the absence of a land title. The main sources for formal credit include commercial banks (government or privately owned), small finance banks (private sector entities servicing small and marginal producers), microfinance institutions (MFIs), cooperatives, self-help groups (SHG) and registered entity of producers that share profits called farmer producer organizations (FPO). Interest rates of loans offered by MFIs and small finance banks vary from 18 to 26% in Odisha and are about 24% in Maharashtra. Loans supplied by commercial banks have lower interest rates but require a collateral and documents that many households cannot provide. They also delay loan approval and disbursement well beyond the planting season when farmers need cash on hand to buy inputs. Loans from SHGs and FPOs do not require collateral for members but are available on rotation and not every planting season. The Indian government offers a collateral-free loan of up to INR 200,000 (about USD 2,250) to farmers called the Kisan Credit Card (KCC) at the maximum interest rate of 7%, however, it is restricted to farmers that own land. This makes majority of our sample ineligible as we show in Table 1. Figure C.1 shows that fewer than 20% of farmers in our control group reported taking the KCC loan in the endline survey.

3.2 Randomization

We draw from the same experiment evaluated in Kramer et al. (2025a) which tests the households' uptake of both credit and insurance, indebtedness, repayment difficulties, costs and revenue from production. In this paper, we evaluate the impacts of the loan intervention arm on household consumption and individual members.[7]

The sample includes 80 villages identified by DER, half of which were located in two blocks of Jajpur district, and the remaining half in four blocks of the Amravati district.[8] Fifteen of the twenty control group villages in Odisha were randomly selected from the control group of an evaluation of a credit and insurance bundle by Kramer et al. (2025b). These villages were not exposed to any credit or credit and insurance bundle offered by DER in the past and had access to market sources of formal and informal credit like

[7] We do not analyze a third intervention arm of this RCT that provided loan and insurance bundles because 15 out of the 20 study villages in that arm were already exposed to the same credit and insurance bundle through an earlier program. The timing of our evaluation does not let us analyze the exogenous and unanticipated impacts of credit access in these villages.

[8] Blocks are administrative sub-divisions of districts used for planning and development in rural areas, sometimes sharing the district name. The sample includes Dashrathpur and Jajpur blocks in Jajpur district, and Amravati, Ashti, Morshi, and Tiosa in Amravati district.

the remaining control group villages in Odisha and Maharahstra that were recruited at the start of the experiment. DER organized meetings in each village September through October in 2022 after consultation with village heads, ensuring farmers were available to attend. DER facilitated a discussion of credit options in the market and did not elicit applications or describe terms of their loan product in these meetings. It also did not include any information that could change social norms around loans, use of mobile phones, financial autonomy and individual welfare. The villages were randomly assigned into treatment and control arms by lottery *after* the conclusion of meetings in all 80 villages. Half the villages (20 per state, 40 in total) were assigned to the treatment arm and the remaining half to control (split equally between the 2 states). A roster of at least 20 farmers was created for each village by eliciting names and contact information from participants of the information session.

3.3 Loan program and Eligibility

DER offered a collateral-free loan to small and marginal farmers by partnering with one commercial and one small finance bank. DER provided these loans in the summer planting season in 2023 and the winter season spanning 2023-24. In this paper, we only analyze the impact of credit provided in the summer planting season as the timing of the endline survey captures the repayment from this season. It also avoids the confounding effects of anticipation of the product. Additionally, most farmers in the surveyed regions did not cultivate during winter. The loan amount ranged between INR 30,000 and 60,000 (USD 330 and 660), varying by the size of the land, and majority of the farmers were offered INR 30,000/ USD 330. It was offered at the beginning of the planting season and farmers repaid the loan in monthly installments by crash or mobile apps.[9] The option to repay through a digital platform may also have increased household members' knowledge and use of digital financial services directly. The program addressed two additional challenges to credit access for these farmers - loan application and processing times. DER agents visited farmers at their field or home and filled out the application on their behalf. The loan was processed within seven business days for successful applicants. Only 3 of the 353 households that borrowed under this program defaulted, yielding a repayment rate of about 99%.

To be eligible for the loan, farmers were required to have attended the meeting organized in their village at the start of the intervention. Eligibility also required residency in the village for at least five years, farming on no more than three acres, and cultivation at least once in the past three years in the borrowing season. Their credit worthiness

[9] The implementing partner reported that farmers expressed preference for a monthly repayment plan over lump sum payment after harvest.

was computed by DER using an innovative approach that utilized remote sensing data on crop productivity, farm area, damage from natural calamities or other factors, soil moisture and irrigation availability, soil nutrients and vegetative growth from the last three years. This weighted index is called KhetScore. Loan applicants were expected to have a KhetScore (or credit score) above 40. Figure C.2 shows that almost all the farmers in the study were above this threshold. Evaluations from established credit bureaus were additionally assessed when available.

Program costs and net returns per farmer

The program included fixed costs such as project management costs, field office and other materials that are listed in Table D.1. Costs that varied by farmer included the calculation of each farmer's credit score, engagement of field staff and discretionary allowances. The loan had 14% interest rate, which resulted in earnings of USD 63-72 per farmer. Subtracting total costs per farmer from the total revenue through interest payments yielded a profit of about USD 13 and 23 per farmer in Odisha and Maharashtra, respectively.

3.4 Data collection and Attrition

Baseline: The survey was conducted between November 2022 and February 2023 by an independent team of enumerators hired and trained by the International Food Policy Research Institute (IFPRI). The enumerators interviewed the farmer who attended the information session. The questionnaire included detailed information about the crops cultivated during the summer and winter agricultural cycles, inputs utilized in cultivation, asset ownership, economic shocks, diet diversity and participation in decision making. Across the 80 villages, the goal was to interview approximately 21 households per village and the baseline data included 1,682 households.

Administrative data: We utilize DER records on the number of households that applied for a loan in the treatment arm, were successful and the loan amount they received during the summer of 2023. We also analyze the credit score estimated by DER in 2023 for each household our sample.

Endline: The endline survey, conducted again by IFPRI with the help of an independent team of enumerators between April and August 2024, successfully re-interviewed 1,431 households from the baseline. The survey was expanded to interview two members of each household: the farmer who had attended the information session before baseline (henceforth, the "panel respondent") and one additional household member, oftentimes their spouse or the most informed available household member. This second member was successfully interviewed in 1,066 households. We expanded the modules from the baseline survey to also measure respondents' access to financial services and their time use (see Appendix B for a description of all variables).

Sample size and Attrition: There is 15% attrition between the baseline and endline surveys and the sample is balanced across the intervention arms (column 1, Appendix Table D.2). Column 2 of the same table shows that households' ex-ante characteristics are generally uncorrelated with attrition, with two exceptions: households with a female panel respondent were more likely to drop out of the sample, and those with an outstanding loan were less likely to do so. We omit two households where the respondents were recorded as below 18 years of age. This gives us a balanced panel of 1,429 households that includes 1,061 other household members.

We report household and panel respondent characteristics from the baseline survey in Table 1. Column 2 and 3 also report averages for treatment and control groups, respectively. Households in both intervention arms were statistically similar on observables before the loan program was implemented. The average respondent interviewed in baseline was middle-aged (about 43 years old) and over one-third were women. Less than half (47%) of the households owned land and, therefore, ineligible for the government's credit program. Borrowing is prevalent - over 40% of the sample took a loan in the 12 months prior to baseline survey, and one in five households had an outstanding loan. Over 90% of the panel respondents had a bank account.

4 Estimation

4.1 Intent-to-Treat

We estimate the intent-to-treat (ITT) effect of the loan intervention on household resource allocation using the ANCOVA model below. This exploits the random assignment of the intervention and a balanced panel of household outcomes and covariates measured using the baseline and endline surveys.[10]

$$Y_{hvb,1} = \alpha_0 + \alpha_1 Y_{hvb,0} + \alpha_2 TREAT_v + \sum_{j=1}^{J} \theta_j \mathbf{X}_{jhvb,0} + \eta_b + \tau_m + \epsilon_{hvb} \tag{1}$$

Equation 1 estimates the effect of the program on household-level outcomes, indexed by h, controlling for their baseline values, $Y_{hvb,0}$. Appendix B describes these variables. Table D.4 shows that outcome variables are balanced between treatment and control groups at baseline, with the exception of total expenditure on hired labor.[11] We control for household covariates described in Table 1 as well as features of soil quality including moisture, nutrient and productivity that were used to determine each plot's credit worthiness in

[10] We do not reweight observations using inverse probability attrition weights as Table D.2 shows that attrition is balanced across intervention arms.

[11] To assess impacts on household labor demand, we use the real wages households paid for male and female labor, which are balanced at baseline.

$\mathbf{X}_{hjvb,0}$.[12] The binary indicator $TREAT_v$ identifies villages v in the treatment (= 1) and control arms (= 0) and coefficient α_2 estimates the ITT effect of the intervention on outcome $Y_{hvb,1}$. We include block fixed effects (η_b) because treatment was stratified by block, an administrative aggregation of villages, and the fixed effects also control for differences in markets and institutions. Any differences in recall due to survey month of the baseline and endline are controlled for using month fixed effects (τ_m). There are 80 villages in the sample and standard errors are clustered by village, which is the level of treatment assignment. We report sharpened q-values (Benjamini et al., 2006; Anderson, 2008) which control the false discovery rate, as a correction for multiple hypothesis testing within each family of outcomes. The family of outcomes are specified in all results tables.

4.2 Local Average Treatment Effect

We analyze the local average treatment effect of the loan on individual-level outcomes using an Instrumental Variables (IV) model. The first stage tests the predictive power of the exogenous assignment of loan treatment, $TREAT_v$, to households h taking up the loan for cultivation in summer of 2023. The endogenous variable of loan take up is defined using administrative data from DER and described in Table D.3.

$$ReceivedLoanProduct_{hv} = \beta_0 + \beta_1 TREAT_v + \sum_{j=1}^{J} \theta_j' \mathbf{Z}_{jihvb,0} + \eta_b + \mu_{hv} \tag{2}$$

Table 2 reports that households were 48.1 percentage points more likely to apply for and received the loan. We refer to treatment households that applied for and received the loan as borrower households. The F statistic is 151, above the conventional threshold of 10, and the instrumental variable explains approximately 30% of the residual variation in loan take-up after partialling out baseline covariates and block fixed effects. The equation below is the Two-Stage-Least-Squares (2SLS) estimation of the instrumented variation of a borrower household on outcome variables.

$$Y_{ihvb} = \gamma_0 + \gamma \widehat{ReceivedLoanProduct}_{hvb} + \sum_{j=1}^{J} \theta_j'' \mathbf{Z}_{jihvb,0} + \phi RespType_{ihvb} + \eta_b + \upsilon_{ihvb} \tag{3}$$

We analyze outcomes for female and male respondents separately using this 2SLS model and test the equality of women's and men's coefficients by comparing estimates

[12] We report estimates conditional on all covariates as there are very small differences in the magnitude of treatment effects when only a subset of covariates identified by post-double selection LASSO method (Belloni et al., 2013) are included.

from the two samples.[13] We control for household characteristics from the baseline survey described in Table 1 in $\mathbf{Z}_{hjvb,0}$ and also control for respondent's age and marital status in the endline survey as this information was only captured for the main respondent at baseline. We also control for whether the respondent is the panel respondent or the other household member ($RespType_{ihvb}$) in equation 3. We include block fixed effects (η_b) and cluster standard errors by village. Estimates of time use also include survey month fixed effects to control for seasonal differences in the respondent's recall of their time allocation. We adjust p-values for multiple hypotheses testing by reporting sharpened q-values.

The model satisfies the following key assumptions of an IV model. First, we satisfy in Table 1 that the treatment assignment is exogenous by verifying that households' observable characteristics are balanced at baseline between the treatment and control arms. Table D.2 shows that attrition of households does not differ by treatment status.[14] Second, the exogenous village level treatment assignment positively affects a household's take up of the loan product (Table 2). Third, the village meetings did not share information that could impact preferences for loans, social norms on financial autonomy and leisure or involvement in decision making. Fourth, we verify implementation fidelity in Table D.3 showing that only households in the treatment arm received the loan product. Finally, any household that met the eligibility criteria listed under section 3.3 could apply for and receive the loan. By this rule, one household's receipt of the loan product did not affect the likelihood of another household in the treatment arm receiving the product.

Since no control household received the loan, the LATE identifies the effect of treatment on the treated and is our main estimand for measuring impact on individual outcomes. We report ITT estimates alongside the LATE estimates of individual-level outcomes in section 5.2 as a robustness check that captures the effect of village assignment to treatment regardless of loan take-up.

5 Results

5.1 Household responses to the loan program

In this section, we provide empirical evidence for Stylized Fact 1. The intervention increased the ownership of productive capital and consumer durables by households in the treatment group relative to the control group. Table 3 shows a 13.8 pp increase in the own-

[13] This is an approximate test given respondents are drawn from the same households and villages, and their estimates are likely correlated.

[14] Nearly all baseline characteristics are uncorrelated with attrition, except for households with a female panel respondent were 6 pp more likely to drop out of the sample and those with an outstanding loan were 6 pp less likely to do so. Treatment coefficient remains insignificant.

ership of farm equipment such as hand tools, animal-drawn plows and non-mechanized irrigation tools in the treatment arm relative to control. Ownership of non-farm business equipment increased by 7.1 pp.[15] Households in the treatment arm were 7.8 pp more likely to own large consumer durables. This includes time-saving kitchen appliances such as refrigerators, televisions, and washing machines as well as other household durables such as sofa, dining tables, chairs, mattresses, cots or beds and computers. Relaxing liquidity constraints allowed households to purchase composite goods that may have been unaffordable without the loan. Because over 99% of loans were repaid, households ultimately financed these purchases from their own resources. We don't find increases in mechanized farm inputs such as tractors, electric pumps, seeder and threshers because even with subsidies from the government, some of these equipments cost at least four times more than the loan amount.

We find relative improvements (about 17%) in the diet quality of the households in the treatment group in Table 4. Panel A measures household diet diversity using a 7 days recall measure of 8 food groups designed by the World Food Programme.[16] We further analyzed whether the individual respondent consumed the same food groups in the last 24 hours in panel B, weighting responses in the same manner as the household food consumption score with frequency 1 (one day). Both women and men's food consumption in the last 24 hours was significantly higher in response to the loan program. Haushofer and Shapiro (2016) find that an unconditional cash transfer in rural Kenya increased households' accumulation of consumer durables and food security. We observe similar patterns following a collateral-free loan. Savings from the lower interest rate of our loan product may have contributed to this pattern, although incomplete data on household borrowing and interest payments prevent us from reliably quantifying their role.

5.2 Impact on household members

Having estimated the household response to the loan, we now report changes in individual level outcomes for female and male household members in our sample. We focus primarily on the interpretation of women's outcomes given the focus of this paper, and discuss how they relate to the impacts on men where relevant.

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[15] Non-farm business equipment includes any equipment used in an enterprise unrelated to farming. The survey listed solar panels used for recharging and sewing machines as examples, but respondents could report other equipment meeting this definition.

[16] The score assigns the following weights (in parentheses) to different food groups: Cereals, grains, roots and tubers (2); legumes/ pulses, nuts and seeds (3); milk and other dairy products (4); meat, fish and eggs (4); vegetables and leaves (1); fruits (1); oil, fat and butter (0.5); sugar and sweets (0.5); condiments (0).

Table 5 shows that the loan product did not impact women's preference for having their own bank account, but they report greater access to resources. Women (men) in borrower households were about 40 pp (34 pp) more likely to deposit funds that were not earnings or transfer payments in their account suggesting increase in their residual income through the resource expansion channel. We find direct impacts on their financial autonomy through debit card use. Women in borrower households were 70 pp more likely to report that the debit card linked to their account was only used by them. Men in borrower households were 14 percentage points more likely to hold an individual account than men in the control group. This is possibly explained by 68% of loan recipients in the treatment arm being men, many of whom may have opened an individual account to receive the loan. Distinguishing men and women as independent sample, the equivalence tests of the estimates are not statistically significant. Men also reported greater likelihood of using their debit card autonomously in response to the loan. Both LATE and ITT estimates are significantly larger in case of women than men and likely driven by differences in the control group mean: fewer than 20% of women in the control arm used their debit card independently, compared with roughly half of the men.

Women were over 80 pp more likely to have knowledge and experience using mobile banking apps in borrower households relative to the control group (Table 6). The increased engagement in digital financial services is a second source of enhanced financial autonomy as these apps increase a user's privacy in transactions. Men in these households also exhibited greater engagement in mobile based transactions than the control group (43 pp more likely to have knowledge and 37 pp use). Women in borrower households reported greater mobile phone ownership (over 13 pp but not statistically significant) relative to the control group. Among women with their own bank account, comprising 83% of the sample, the difference in mobile phone ownership was statistically significant (24 pp, LATE estimate and 12 pp, ITT). The effects on knowledge and use of mobile banking apps were also larger for these women than for the full sample. We treat these estimates as descriptive since both individual-level bank account and smartphone ownership were measured at the endline. The intervention did not appear to explicitly change household take-up of smartphones as about 74% of households owned at least one smartphone at baseline, and there were no changes on the extensive margin between the treatment and control groups over time. Equivalence tests of the coefficients of knowledge and use of mobile banking apps show that the treatment effects for women are significantly larger than those for men, again owing to a smaller control group mean for women. This indicates that the loan product narrowed gender gaps in knowledge and use of mobile banking. Finally, we show that the positive impacts on financial autonomy are consistent across both states in our sample in Tables D.12 and D.13.

The positive impact of the loan product on women's uptake of digital financial services is noteworthy, since women's internet access have been persistently low in India owing to rigid gender norms (Barboni et al., 2018). The results on increased engagement in mobile apps for banking, largely explained by the expansion of women's access to resources in borrower households, contrast with the findings of Barboni et al. (2024) where smartphone provision to rural women in Chhattisgarh led to no impact on women's knowledge or use of mobile banking. Instead, they found that 40% of the women transferred their phone to another household member, highlighting that targeted asset provision doesn't guarantee resource control.

Time use

We asked women their time allocation across different production and consumption activities listed in Table B.1 to test implications of the loan on individual labor supply. Table 7 shows that women in borrower households spent about 0.94 (53%) more hours on personal activities such as visiting friends, watching TV etc., than the control group (mean = 1.68 hours) during the busiest day of the past agricultural season. This result suggests an income response to increase in residual income as leisure is a normal good in our theoretical expectations. Time allocations across activities do not add upto 24 because respondents were asked the time they spent on activities in Table B.1 conditional on them reporting they participated in that activity. Appendix B explains the structure of the survey module.

Women in borrower households allocated about 2.3 hours less on farm activities than women in the control group (mean = 3.1 hours) on the busiest day in the agricultural season, however, this reduction is not representative of the entire season. Table D.7 shows that the number of days worked on the farm in the past agricultural season is not significantly different for female household members between the treatment and control group as recalled by the panel respondent.[17] Women's reduction of hours to farm work on the busiest day is consistent in the ITT estimates (1.9 hours in treatment versus 3.1 hours in control group). If the panel respondent recalls household members' labor supply correctly, the results imply that although women significantly reduced the number of hours they allocated to farming activities in a day, their contribution in terms of the number of days worked over the season did not differ significantly between the treatment and control groups.

[17] We observe seasonal labor supply, farm income, yield, and household input use in both the baseline and endline surveys, therefore, we estimate effects on these outcomes using the ANCOVA model, which compares treatment and control groups at endline while controlling for baseline values of the outcome.

Despite women allocating fewer hours to farm activities in the treatment group relative to the control group, Table D.8 shows there were no impacts on real farm income, total quantity harvested and yield. One explanation for the reduction in farm hours in a day is that households' ownership of non-mechanized farm equipment allowed women to work more productively. As described in section 3.1, women are primarily responsible for manual tasks such as transplanting, weeding and picking while men operate machinery and perform market-facing activities. Additionally, in Table D.9, we report no significant differences in households' payment of real wages to male or female hired labor. There were also no relative changes in households' expenditure on *other inputs*[18] such as irrigation, purchase of seeds and fertilizers or renting of machinery and land (Table D.10) that may have enhanced the demand for farm labor.[19] Men's time allocation to farm and non-farm production on the busiest day of the agricultural season remained statistically similar between borrower and control group households, lending support to the productivity enhancing channel for women.

Women also reported a reduction in time spent on non-agricultural work such as self employment or wage employment on both the busiest day of the agricultural season. Since ITT estimates of differences between the treatment and control groups are consistent, we explore the corresponding accumulation of business equipment by households in the treatment villages relative to the control as a possible productivity-enhancing explanation, however, we do not have corresponding income or output measures to verify. The large point estimates of reduction in hours worked indicate a significant reduction of women's participation in non-agricultural work on the extensive margin. This may be the result of households relying less on small businesses to meet their budgetary needs after receiving the loan. Table D.6 shows that among women who did participate in the activity, in borrower households, they allocated significantly more hours towards non-agricultural work (2.1 hours) and less towards domestic work (0.95 hours) in the day before the survey. We interpret these estimates cautiously because the sharpened q value is about 0.18 for these outcomes, which allows 18% of false rejections in each set (busiest day and day prior to survey).

We examine whether women's perceptions of gender roles bias their time-use reports as a robustness check. In this test, we assume that women who have their own bank account are less affected by any social biases around women's economic participation and may provide more accurate reports of their time allocation than women who don't. To

[18] This result is robust to households' use of *any formal loans*. Table D.11 shows that the treatment group did not utilize loans to finance spending on seeds, fertilizers, labor or renting machinery.

[19] These results apply only to the sample analyzed in this paper and are not conclusive evidence on the impact of the program overall. We analyze a balanced panel of households in the loan treatment and control villages as attrition is balanced across arms.

the extent that this captures women's insulation from social norms, we find no heterogeneous effects of women's time allocation to farm or non-agricultural work by independent bank account ownership in Tables D.15 and D.16.

Men's economic participation remains statistically similar between the borrower households and the control group. The ITT estimates of all time use results are consistent with the LATE estimates for both women and men. Time allocation on the day before the survey follows the same patterns as on the busiest day of the agricultural season, with one exception. Women in borrower households report significantly fewer hours of domestic chores than the control group on the day before the survey, whereas the corresponding estimate for the busiest day is not statistically significant.

We examine women's time allocations by state given differences in the crops planted and production processes across the two states. Tables D.17 and D.18 show that the reductions in economic participation are driven by households in Maharashtra and greater allocation to leisure by Odisha. Households' report of farm labor supply by members over the season, expenditure on labor (real wages and total real spending) and real income from farming are not significantly different between the treatment and control groups in either Odisha or Maharashtra (Tables D.19-D.21), validating that any reduction in farm labor on the busiest day in the agricultural season is not consistent over the season. The pattern of women's reduction in non-farm work in Maharashtra and greater time allocation to leisure in Odisha is possibly the result of differences in social norms between the two states. Maharashtra is systematically more conservative in gender norms than Odisha. It matches the findings in El-Enbaby et al. (2025) where women in rural Egypt reduced non-farm work but not their participation in agriculture in response to a cash transfer, consistent with greater social acceptance of women working for pay in agriculture than non-farm work in this setting.

Decision-making:
Finally, we investigate if there are potential impacts on individual residual income through changes in bargaining power measured as women's involvement in decision-making. Figure 2 reports LATE estimates for the share of decisions women participated in over the last 12 months, along with category-wise participation. The loan product did not affect the share of decision categories in which women or men participated, nor their involvement in any individual category on the extensive margin. The one exception is farming decisions, where women in borrower households were 21 pp less likely to participate than the control group (mean = 0.68), though the adjusted p-value exceeds conventional thresholds.

For decisions made jointly with their spouse (Table D.23), women in borrower households are more likely to participate in livestock and poultry management and household consumption, however, the sharpened q-value is 0.153, therefore, we interpret these results cautiously. There are no differences in men's reports of joint decision making with their spouse between borrower households and control group. In panel C, we restrict the data to approximate households where spouses were interviewed[20] and find that the the loan product did not differentially impact whether both spouses report joint decision making consistently. We therefore treat the overall pattern as one of no *aggregate* impacts on intrahousehold bargaining power.

The empirical evidence of targeted programs on women's decision making is mixed. Cash transfer programs have had success (de Brauw et al., 2014; Bergolo and Galván, 2018) while social protection programs and bank account ownership did not (Roy et al., 2015; Field et al., 2021; Chauhan, 2025). Other financial inclusion interventions such as microfinance, savings groups, business loans and grants have had limited impact on women's decision making ability (Banerjee et al., 2015; Karlan et al., 2017; Angelucci et al., 2015). Women's initial bargaining power in the household drives differences in program impacts in some contexts (Hidrobo and Fernald, 2013; El-Enbaby et al., 2025). We, therefore, investigate this mechanism in the next section.

6 Mechanisms

In this section, we examine potential factors that could impact individual welfare such as ex-ante measures of households' wealth, socio-economic identity and spousal coordination.[21]

Ex-ante wealth and socio-economic identity: We construct a wealth index using principal component analysis of the assets owned by a household at baseline and classify all households in our sample by quintile of this index. Interacting this categorical variable with the instrumental variable for whether households took up the loan shows no significant correspondence with women's or men's financial inclusion as measured through bank account ownership, use of debit card, mobile phone ownership and engagement with mobile apps for banking. Table D.26 shows that in wealthier households that borrowed

[20] These are households where two members were interviewed successfully in the endline survey. We further assume that the respondents are spouses in cases where either the woman interviewed was upto 19 years younger than the man (95th percentile of the distribution of age difference) or the man was upto 6 years younger than the woman (25th percentile).

[21] We estimate the heterogeneous treatment effects using 2SLS. Loan take-up and its interaction with a baseline covariate are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same baseline covariate.

under this program, women spent less time on household purchases and domestic chores than in the control group on the busiest day of the agricultural season. These results are statistically significant with 10.7% chance of false rejection after adjusting for multiple hypotheses testing. Men in wealthier borrower households, also spent less time on household purchases, and allocated more time to personal activities such as leisure. The heterogeneity in time allocation by women does not persist in the day prior to the survey. However, Table D.27 shows that wealth drives fewer hours spent by men in livestock and poultry management and more hours in personal activities on the day prior to the survey. Table D.28 reports no heterogeneous effects by asset quintile on women's and men's involvement in decision-making with the exception of men in wealthier households relatively less likely to participate in household consumption decisions, consistent with their time allocations on the busiest day of the agricultural season to this activity.

Households belonging to Scheduled Caste (SC) and Scheduled Tribes (ST) are historically more disadvantaged, therefore, we investigate if the loan impacted SC/ ST households differently from the rest in Tables D.29-D.33. We find no heterogeneous effects in the individual-level outcomes with the exception of women in borrower households that identified as SC/ ST were more likely to know how to use mobile apps for banking, relative to the control group.

Historically, households belonging to Scheduled Castes (SC) and Scheduled Tribes (ST) have been more disadvantaged, so we test whether the loan affected them differently in Tables D.29-D.33. We find no heterogeneous effects on individual-level outcomes, with one exception: among women, the loan's effect on knowledge and use of mobile banking were larger.

Ex-ante coordination between spouses: We estimate spousal coordination within a household using a continuous variable of the share of decisions on production and consumption activities that the panel respondent made with their spouse before the introduction of the loan program. While the loan did not change women's involvement in decision-making on average, it had a larger effect on decision-making in households where spouses were already coordinating more before the intervention. Table D.34 reports that women were more likely to be involved in decisions on non-agricultural work, household and private consumption, domestic chores and personal activities. There were positive heterogeneous effects on men's involvement in decisions on non-agricultural activities.

Greater ex-ante spousal coordination encouraged more time allocation to non-agricultural activities and household purchases for both women and men in borrower households rel-

ative to the control group during the busiest day in the past agricultural season (Table D.35). Women in these households also allocated relatively more time to domestic chores. In the day prior to survey, there were no heterogeneous effects on men's time allocation while women reported more time allocation towards non-agricultural work and domestic chores (Table D.36).

Table D.37 reports that ex-ante spousal coordination corresponded negatively with women's individual account ownership in borrower households, however, there were no differential trends in autonomous debit card usage or knowledge and use of mobile banking apps (Table D.38).

7 Selection into program

In this section, we distinguish effects on women's agency by a household's selection into the loan program. Within the randomly selected treatment villages, households that attended the information session were offered the loan. These 702 households can be further split into 3 categories based on their loan application status - (i) applied and received the loan (352), (ii) applied but were unsuccessful (86) and (iii) did not apply for the loan (310). In Appendix Table D.39, we show that borrower households (category i) were statistically similar in observable characteristics with households in the control arm. The unsuccessful loan applicants (category ii) had higher prevalence of literacy and bank account ownership than households in the control arm. However, these differences may be attributed to noise/ high sampling variability from a small sample size of 86. The households that did not apply for the loan (category iii) were less likely to have a bank account or have taken a loan in the year before the intervention than the control group. Half the households belonged to SC/ST. More than half of them were in the lowest income group captured in the survey (< USD 900 per year) and had lower schooling outcomes (35% of the panel respondents had completed secondary schooling vs. 50% in the control group).

Given these systematic differences within groups in the treatment arm, we follow Crépon et al. (2015) and estimate a propensity score of borrowing based on observable characteristics in the baseline survey (Figure C.3). We re-weight the LATE by each household's ex-ante predicted probability of loan receipt. The positive impacts on women's independent debit card use and both women and men's knowledge and use of mobile apps for banking are consistent in the re-weighted estimates reported in Tables D.40 and D.41. Men report a greater likelihood to share their debit card with other family members in borrower households than the control group. Women's time allocation

of relatively more hours to personal activities and fewer to non-agricultural and farm work on the busiest day in the agricultural season were also consistent in Table D.42. Men reported spending fewer hours on livestock and poultry and more hours on domestic chores and personal activities. There were no aggregate differences in women or men's involvement in decision making with the exception of women reporting less likelihood to participate in decisions about farming and more in decisions on domestic work, however, adjustments for multiple hypotheses testing report very large false rejection rates.

The effect sizes of financial inclusion indicators are smaller than results in section 5.2 indicating that borrower households experienced larger effects than the average likely borrower would.

8 Conclusion

This paper investigates whether an economic intervention at the household level can positively impact women's access to resources. We study this in the context of a collateral free loan to farming households in India. In contexts where household members participate in join production, such as agriculture in India, utilization of credit is unlikely to vary based on who is allocated the loan but can benefit individual members in two ways. First, if the loan raises the household's joint income in the short term and directly increases the consumption of composite goods, such as productive capital and consumer durables that are shared or privately consumed, individual members' utility will increase from their consumption of composite goods. Second, individual members' residual income may change through a resource expansion channel or bargaining power channel which changes their residual income, resulting in a change in their private consumption and labor supply.

We study this using an RCT where a collateral-free loan was provided to small and marginal farmers in 40 of the 80 study villages in India. We combine the randomized loan provision with a balanced panel of 1,429 households interviewed in baseline and endline surveys. Households in both treatment and control groups were balanced on observable characteristics at baseline. We exploit the baseline and endline rounds to analyze the intent-to-treat effects of the loan program on household resource allocation using an ANCOVA model. We analyze the individual level outcomes collected in the endline survey and use a 2SLS model to estimate the local average treatment effect of the loan product on individual financial inclusion, time use and involvement in decision making. This model instruments the exogenous village-level treatment assignment for borrower households.

We report three main results. First, households increased their ownership of productive equipment for farm work (14 pp) and for non-farm business (7 pp), as well as large

consumer durables (8 pp), and their diet diversity improved by 17%. Second, women in borrower households report greater access to money. They were 40 pp more likely to receive deposits in their bank accounts two to three times a month. Alongside this expansion in resources, they exercised greater financial autonomy. They were 70 pp more likely to use their debit card on their own and over 80 pp more likely to know about and use mobile banking services. Third, women's time allocation shifted. On the busiest day of the agricultural season, women in the treatment group spent more time on leisure (2.6 versus 1.7 hours) and less time on farm work (0.8 versus 3.1 hours), although their labor supply over the season does not change. Farm income and yield are also unchanged, so the equipment may have allowed women to work more productively. Women also spend less time on non-agricultural activities such as business or wage employment, which may reflect lower demand for non-farm income after the loan. This decline is concentrated in Maharashtra, suggesting that social norms may also shape women's response. Finally, the loan did not change women's involvement in decisions on production or other activities, however, the loan reinforced existing cooperative household arrangements. Women in households with greater ex-ante coordination among spouses on decisions, were more involved in decision-making.

For policymakers, this evidence highlights the downstream welfare effects of agricultural support programs that aren't accounted for when evaluating their benefits. Women in rural agrarian households typically face limited financial access (Demirguc-Kunt et al., 2022) and there is a critical need for cost-effective, scalable programs that improve women's access to resources. By positively impacting women's independent use of financial services, such interventions extend benefits beyond farm productivity and income.

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9 Figure

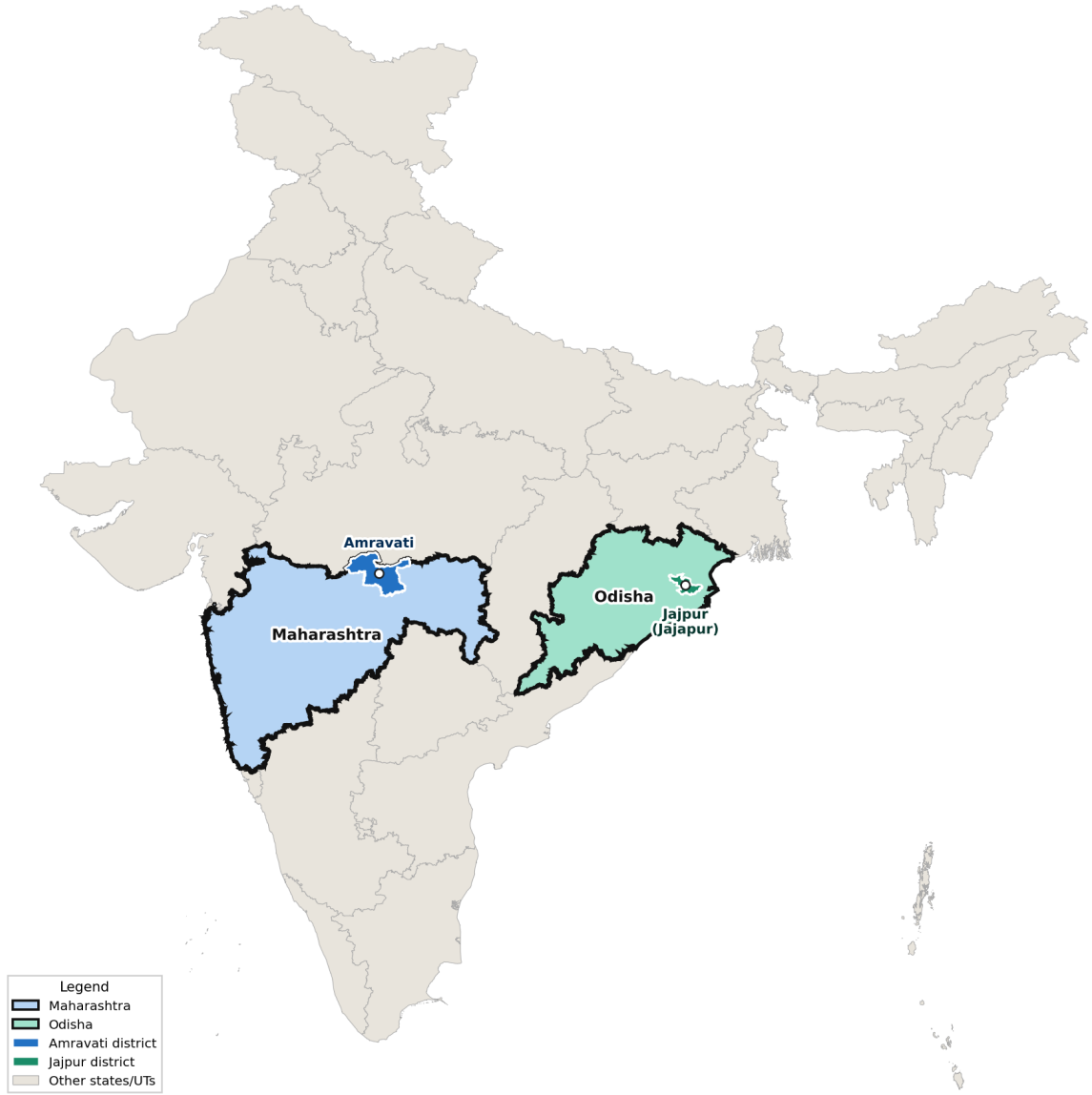


Figure 1: Study location

Notes: The figure indicates the two districts in India where the program was implemented - Amravati in Mahrahstra (blue) and Jajpur in Odisha (green).

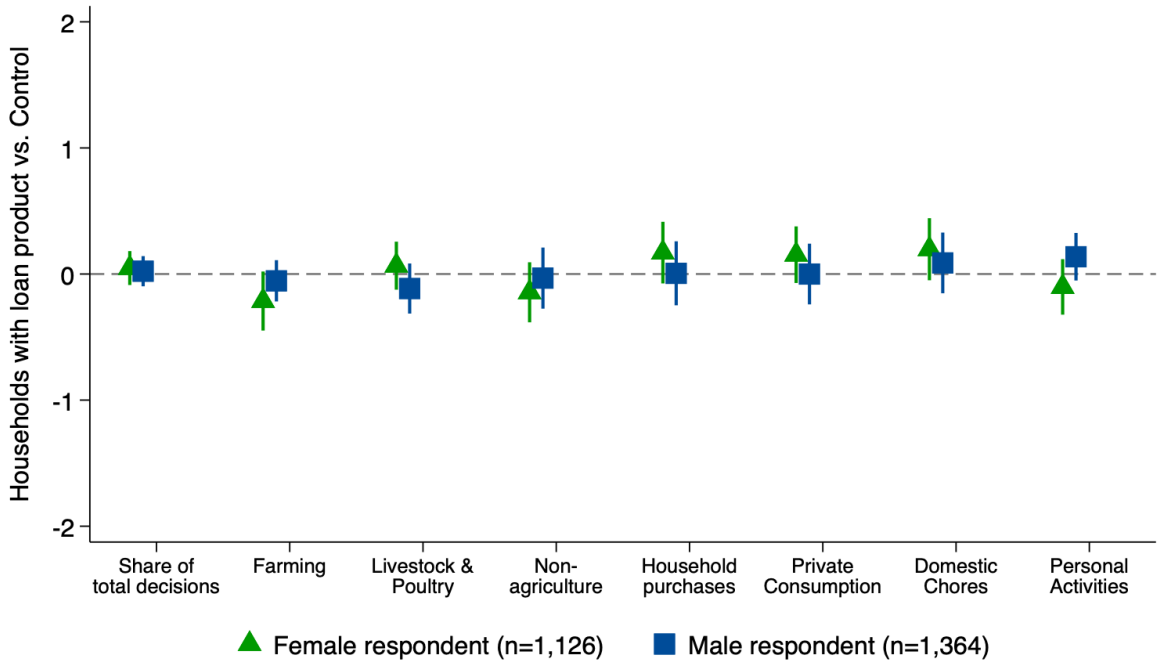


Figure 2: LATE: Loan intervention and involvement in decision-making

Notes: The figure the local average treatment effect of the loan on women and men's involvement in decision making on production and non-production activities (described in Table B.1). Sharpened q values report statistical significance of coefficients after correcting for multiple hypothesis testing. $*$ q$<$ 0.1, $**$ q$<$ .05, $***$ q$<$ .01. The p-value of an equivalence test is reported if the coefficients of men and women differ significantly from each other.

10 Tables

Table 1: Baseline covariates by intervention arm

	(1) All	(2) Loan	(3) Control	F-test for balance across all groups	(2)-(3) Difference
Respondent's age	43.301	43.824	42.727	1.639	1.097
	(-0.447)	(-0.723)	(-0.469)	0.204	
Respondent is female	0.367	0.37	0.363	0.008	0.008
	(-0.042)	(-0.052)	(-0.067)	0.929	
Respondent is married	0.934	0.941	0.927	0.484	0.015
	(-0.01)	(-0.011)	(-0.018)	0.489	
Can read and write	0.875	0.872	0.88	0.065	-0.008
	(-0.016)	(-0.021)	(-0.023)	0.8	
Completed at least primary school	0.788	0.769	0.809	0.889	-0.04
	(-0.022)	(-0.03)	(-0.031)	0.349	
Completed at least	0.457	0.413	0.505	2.516	-0.092
secondary school	(-0.029)	(-0.039)	(-0.044)	0.117	
Respondent belongs	0.358	0.39	0.323	0.991	0.067
to SC/ST category	(-0.035)	(-0.05)	(-0.046)	0.322	
Annual Household	0.474	0.511	0.433	0.82	0.078
Income(<90,000)	(-0.043)	(-0.059)	(-0.063)	0.368	
Annual Household Income	0.237	0.235	0.239	0.008	-0.004
(90,000 - 120,000)	(-0.023)	(-0.032)	(-0.034)	0.93	
Annual Household	0.289	0.254	0.327	0.978	-0.073
Income(>120,000)	(-0.038)	(-0.053)	(-0.053)	0.326	
Owns land	0.479	0.493	0.464	0.16	0.029
	(-0.037)	(-0.05)	(-0.054)	0.69	
Asset quintile	2.986	3.043	2.924	0.273	0.119
	(-0.114)	(-0.155)	(-0.169)	0.602	
Borrowed money last	0.439	0.418	0.463	0.428	-0.044
12 months	(-0.034)	(-0.046)	(-0.05)	0.515	
Has outstanding loan	0.212	0.225	0.198	0.141	0.026
	(-0.035)	(-0.051)	(-0.048)	0.708	
Respondent has bank	0.933	0.929	0.937	0.158	-0.008
account	(-0.01)	(-0.013)	(-0.014)	0.692	
F-test of joint significance (F-stat)				0.725	
N	1429	748	681		

Notes: The table reports averages of respondent and household characteristics from the baseline survey by intervention arm. Column 1 reports pooled estimates for the full sample. Column 2 reports the averages for the loan treatment arm and column 3 for control arm. The fourth column reports the f test for whether each variable is balanced across arms and the last column conducts a pairwise t-test of the mean difference between columns 2 and 3. "SC/ST" is abbreviation for Scheduled Caste/ Scheduled Tribe. Standard errors are clustered at the level of treatment assignment (village) and reported below mean values for each variable. The F-test of joint significance omits the highest household income category to test for variance across all variables. The number of observations are included in the last row. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table 2: Instrumental Variables: First stage

	(1)	(2)	(3)
Treatment: Loan	0.471***	0.472***	0.481***
	(0.046)	(0.043)	(0.039)
Constant	0.000	0.006	0.245
	(0.000)	(0.103)	(0.172)
Observations	1429	1429	1429
Baseline covariates	No	Yes	Yes
Block fixed effects	No	No	Yes
F statistic (instrument)	105.939	121.435	151.354
Partial R-squared	0.298	0.294	0.303

Notes: The table reports ordinary least squares estimation of village's assignment to treatment on household receiving loan product for summer cultivation in 2023. This estimation uses the endline survey only. Columns 2 and 3 include controls measured at baseline and described in Table 1. Column 3 includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table 3: ITT: Loan intervention and household asset ownership

	Non-mechanized farm equipment	Mechanized farm equipment	Business equipment	House and structures	Large consumer durables	Small consumer durables
	(1)	(2)	(3)	(4)	(5)	(6)
ITT: Village	0.138**	-0.001	0.072**	-0.034	0.078***	-0.071
assigned to	(0.067)	(0.041)	(0.029)	(0.026)	(0.029)	(0.057)
treatment (β)	{0.059}	{0.341}	{0.052}	{0.146}	{0.052}	{0.146}
Mean dependent variable (Baseline)	0.241	0.275	0.132	0.742	0.812	0.527
LATE: Received	0.258*	0.002	0.135**	-0.106*	0.153**	-0.146
loan product (γ)	(0.136)	(0.082)	(0.057)	(0.056)	(0.060)	(0.113)
	{0.068}	{0.319}	{0.068}	{0.068}	{0.068}	{0.104}
Control group mean	0.368	0.444	0.194	0.966	0.834	0.633
Observations	1361	1351	1343	1365	1354	1346
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports the ITT and LATE estimates of the loan product on household ownership of assets (listed as column titles). The dependent variables are binary and constructed from survey questions on whether household owns the following: Column 1 includes non-mechanized equipment for farming such as hand tools, animal-drawn plows and non-mechanized irrigation tools. Column 2 includes mechanized farm equipment such as electric pump-set, diesel pump-set, tractor, seeder, power-tiller, thresher, combine harvester etc. Column 3 includes non-farm business equipment such as sewing machines and solar panels. Column 4 includes large consumer durables such as refrigerator, TV, sofa, dining table, chair, mattress, cot or bed, computer, washing machine. Column 5 includes small consumer durables (radio or transistor, pressure cooker, an inverter, electric fan, watch or clock, landline phone). The model includes outcome variable and covariates from the baseline survey as regressors and block fixed effects.Sharpened q-values are reported in curly braces. A family consists of the three outcomes within a panel, and LATE and ITT estimates are treated as separate families. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table 4: ITT: Food Consumption Score (FCS)

	(1)	
Panel A: Household food consumption score		
ITT estimate	5.611*	
	(2.845)	
Observations	701	
Mean dependent variable (Baseline)	33.799	
Covariates included	Yes	
Block fixed effects	Yes	
Panel B: Individual food consumption score (24 hrs recall)		
	Women	Men
	(1)	(2)
LATE estimate	1.966*	2.829*
	(1.183)	(1.552)
Observations	395	342
Control group mean	10.661	10.661
Covariates included	Yes	Yes
Block fixed effects	Yes	Yes

Notes: The table reports the effect of loan treatment on food consumption score in panel A (ITT estimates), and the effect of a household receiving the loan product on individual food consumption score in last 24 hours in panel B (LATE estimates). Panel A uses an ANCOVA model where outcome variables and covariates measured at baseline are used as regressors in the estimation. Panel B uses treatment assignment as an exogenous instrument for the household's receipt of loan product. All models control for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table 5: LATE and ITT: Individual bank account and debit card

	Own bank account (1)	Deposited money (2)	Withdrew money (3)	Received money (4)	Use debit card alone (5)	Share debit card (6)
Panel A: Women						
LATE: Received loan product (γ_F)	-0.021	0.399***	0.442***	0.244**	0.709***	-0.133
	(0.097)	(0.108)	(0.103)	(0.102)	(0.123)	(0.097)
	{0.268}	{0.001}	{0.001}	{0.015}	{0.001}	{0.076}
ITT: Village assigned to treatment (β_F)	-0.010	0.197***	0.218***	0.120**	0.350***	-0.066
	(0.048)	(0.051)	(0.047)	(0.049)	(0.055)	(0.048)
	{0.265}	{0.001}	{0.001}	{0.013}	{0.001}	{0.075}
First-stage F	111.614	111.614	111.614	111.614	111.614	111.614
Observations	1126	1126	1126	1126	1126	1126
Control group mean	0.857	0.374	0.421	0.362	0.178	0.309
Panel B: Men						
LATE: Received loan product (γ_M)	0.140*	0.347***	0.272**	0.274**	0.249***	0.033
	(0.076)	(0.123)	(0.116)	(0.125)	(0.091)	(0.087)
	{0.045}	{0.023}	{0.029}	{0.033}	{0.023}	{0.133}
ITT: Village assigned to treatment (β_M)	0.066*	0.163***	0.128**	0.129**	0.117**	0.016
	(0.036)	(0.059)	(0.057)	(0.061)	(0.046)	(0.040)
	{0.060}	{0.040}	{0.040}	{0.041}	{0.040}	{0.132}
First-stage F	147.900	147.900	147.900	147.900	147.900	147.900
Observations	1364	1364	1364	1364	1364	1364
Control group mean	0.836	0.303	0.400	0.217	0.520	0.227
$\gamma_F - \gamma_M$	-0.162	0.053	0.170	-0.030	0.460	-0.166
p-value of test ($\gamma_F = \gamma_M$	0.191	0.748	0.272	0.851	0.003	0.203
$\beta_F - \beta_M$	-0.076	0.034	0.090	-0.008	0.233	-0.081
p-value of test ($\beta_F = \beta_M$)	0.204	0.659	0.223	0.916	0.001	0.194
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE and ITT estimates of the loan product on household members' bank account ownership and debit card use. LATE estimates are from a 2SLS model using village assignment to treatment as an instrument for a household's receipt of the loan product. ITT estimates are from ordinary least squares regressions of the outcome on village assignment to treatment, estimated on the same sample as the corresponding LATE. The dependent variables (binary) are listed as column titles. Panel A restricts the sample to women in the endline survey and Panel B to men. All regressions control for the household's demographic and economic characteristics from the baseline survey and include block fixed effects. The first-stage F is the F statistic for regressing village assignment on household's receipt of the loan product, conditional on the controls and fixed effects. Standard errors in all models are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q-values are reported in curly braces. A family consists of the three outcomes within a panel, and LATE and ITT estimates are treated as separate families. The test of equality of coefficients between women and men assumes independence between the two samples. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table 6: LATE and ITT: Mobile phone and digital financial services

	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)
Panel A: Women			
LATE: Received	0.138	0.890***	0.809***
loan product (γ_F)	(0.120)	(0.125)	(0.112)
	{0.094}	{0.001}	{0.001}
ITT: Village	0.070	0.439***	0.399***
assigned to	(0.061)	(0.054)	(0.053)
treatment (β_F)	{0.094}	{0.001}	{0.001}
First-stage F	97.878	111.614	111.614
Observations	794	1126	1126
Control group mean	0.300	0.219	0.184
Panel B: Men			
LATE: Received	0.082	0.433***	0.374***
loan product (γ_M)	(0.104)	(0.082)	(0.083)
	{0.169}	{0.001}	{0.001}
ITT: Village	0.038	0.203***	0.176***
assigned to	(0.048)	(0.038)	(0.043)
treatment (β_M)	{0.169}	{0.001}	{0.001}
First-stage F	162.379	147.900	147.900
Observations	1007	1364	1364
Control group mean	0.717	0.609	0.582
LATE difference ($\gamma_F - \gamma_M$)	0.056	0.457	0.435
p-value of test ($\gamma_F = \gamma_M$)	0.725	0.002	0.002
ITT difference ($\beta_F - \beta_M$)	0.032	0.236	0.224
p-value of test ($\beta_F = \beta_M$)	0.684	0.000	0.001
Baseline controls	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes

Notes: The table reports LATE and ITT estimates of the loan product on household members' phone ownership and use of mobile app banking. LATE estimates are from a 2SLS model using village assignment to treatment as an instrument for a household's receipt of the loan product. ITT estimates are from ordinary least squares regressions of the outcome on village assignment to treatment, estimated on the same sample as the corresponding LATE. The dependent variables (binary) are listed as column titles. Panel A restricts the sample to women in the endline survey and Panel B to men. All regressions control for the household's demographic and economic characteristics from the baseline survey and include block fixed effects. Sample sizes differ as non-response to the mobile ownership questions are marked missing. The first-stage F is the F statistic for regressing village assignment on household's receipt of the loan product, conditional on the controls and fixed effects. Standard errors in all models are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q-values are reported in curly braces. A family consists of the three outcomes within a panel, and LATE and ITT estimates are treated as separate families. The test of equality of coefficients between women and men assumes independence between the two samples. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table 7: LATE and ITT: Time use (peak time in agricultural season)

	Personal activity (1)	Non-agriculture (2)	Farming (3)	Livestock & Poultry (4)	Household purchases (5)	Domestic work (6)
Panel A: Women						
LATE: Received	0.939**	-2.171***	-2.286**	-0.195	-0.084	-0.439
loan product (γ_F)	(0.409)	(0.570)	(1.081)	(0.186)	(0.261)	(0.471)
	{0.067}	{0.002}	{0.067}	{0.272}	{0.597}	{0.272}
ITT: Village	0.433***	-1.126***	-1.186**	-0.101	-0.044	-0.204
assigned to	(0.146)	(0.295)	(0.579)	(0.096)	(0.137)	(0.224)
treatment (β_F)	{0.012}	{0.002}	{0.063}	{0.282}	{0.602}	{0.282}
First-stage F	31.143	120.118	120.118	120.118	120.118	35.821
Observations	443	1119	1119	1119	1119	453
Control group mean	1.676	1.323	3.130	0.356	0.677	3.188
Panel B: Men						
LATE: Received	0.256	0.079	0.495	-0.283	-0.174	0.964**
loan product (γ_M)	(0.197)	(0.458)	(0.797)	(0.227)	(0.207)	(0.457)
	{0.566}	{0.765}	{0.765}	{0.566}	{0.765}	{0.312}
ITT: Village	0.123	0.040	0.250	-0.143	-0.088	0.448**
assigned to	(0.091)	(0.231)	(0.398)	(0.116)	(0.106)	(0.191)
treatment (β_M)	{0.582}	{0.791}	{0.791}	{0.582}	{0.791}	{0.157}
First-stage F	114.137	170.314	170.314	170.314	170.314	84.039
Observations	528	1353	1353	1353	1353	311
Control group mean	1.602	1.247	4.144	0.489	0.677	1.611
LATE difference ($\gamma_F - \gamma_M$)	0.684	-2.251	-2.781	0.087	0.089	-1.403
p-value of test ($\gamma_F = \gamma_M$)	0.132	0.002	0.038	0.766	0.788	0.033
ITT difference ($\beta_F - \beta_M$)	0.310	-1.166	-1.436	0.042	0.044	-0.652
p-value of test ($\beta_F = \beta_M$)	0.072	0.002	0.041	0.782	0.800	0.027
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Interview month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE and ITT estimates of the loan product on household members' time use during the peak agricultural season, measured in hours. LATE estimates are from a 2SLS model using village assignment to treatment as an instrument for a household's receipt of the loan product. ITT estimates are from ordinary least squares regressions of the outcome on village assignment to treatment, estimated on the same sample as the corresponding LATE. The dependent variables are listed as column titles (see Table B.1 for description). Panel A restricts the sample to women in the endline survey and Panel B to men. All regressions control for the household's demographic and economic characteristics from the baseline survey and include block and interview month fixed effects. The first-stage F is the F statistic for regressing village assignment on household's receipt of the loan product, conditional on the controls and fixed effects. Standard errors in all models are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q-values are reported in curly braces. A family consists of the six outcomes within a panel, and LATE and ITT estimates are treated as separate families. The test of equality of coefficients between women and men assumes independence between the two samples. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table 8: LATE and ITT: Time use (hours yesterday)

	Personal activity (1)	Non-agriculture (2)	Farming (3)	Livestock & Poultry (4)	Household purchases (5)	Domestic work (6)
Panel A: Women						
LATE: Received loan product (γ_F)	0.856**	-0.958*	-2.149**	-0.235	-0.378	-0.945*
	(0.372)	(0.551)	(0.894)	(0.219)	(0.559)	(0.544)
	{0.082}	{0.096}	{0.082}	{0.151}	{0.201}	{0.096}
ITT: Village assigned to treatment (β_F)	0.392***	-0.491*	-1.103**	-0.121	-0.194	-0.450*
	(0.141)	(0.279)	(0.473)	(0.112)	(0.293)	(0.239)
	{0.046}	{0.089}	{0.060}	{0.140}	{0.205}	{0.089}
First-stage F	34.106	123.539	123.539	123.539	123.539	41.673
Observations	408	1081	1081	1081	1081	416
Control group mean	1.788	0.953	2.244	0.402	1.326	3.528
Panel B: Men						
LATE: Received loan product (γ_M)	0.004	0.461	0.391	-0.343	-0.081	-0.093
	(0.228)	(0.413)	(0.654)	(0.271)	(0.578)	(0.419)
	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}
ITT: Village assigned to treatment (β_M)	0.002	0.232	0.197	-0.173	-0.041	-0.043
	(0.113)	(0.208)	(0.329)	(0.136)	(0.292)	(0.195)
	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}
First-stage F	126.576	167.061	167.061	167.061	167.061	82.892
Observations	536	1345	1345	1345	1345	321
Control group mean	2.016	1.272	2.842	0.521	1.443	1.866
LATE difference ($\gamma_F - \gamma_M$)	0.852	-1.419	-2.540	0.108	-0.297	-0.851
p-value of test ($\gamma_F = \gamma_M$)	0.051	0.039	0.022	0.757	0.712	0.215
ITT difference ($\beta_F - \beta_M$)	0.389	-0.723	-1.300	0.052	-0.153	-0.407
p-value of test ($\beta_F = \beta_M$)	0.031	0.037	0.024	0.768	0.712	0.187
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Interview month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE and ITT estimates of the loan product on household members' time use in the day before the survey, measured in hours. LATE estimates are from a 2SLS model using village assignment to treatment as an instrument for a household's receipt of the loan product. ITT estimates are from ordinary least squares regressions of the outcome on village assignment to treatment, estimated on the same sample as the corresponding LATE. The dependent variables are listed as column titles (see Table B.1 for description). Panel A restricts the sample to women in the endline survey and Panel B to men. All regressions control for the household's demographic and economic characteristics from the baseline survey and include block and interview month fixed effects. The first-stage F is the F statistic for regressing village assignment on household's receipt of the loan product, conditional on the controls and fixed effects. Standard errors in all models are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q-values are reported in curly braces. A family consists of the six outcomes within a panel, and LATE and ITT estimates are treated as separate families. The test of equality of coefficients between women and men assumes independence between the two samples. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

11 Online Appendix A: Theoretical Model

11.1 Set-up

Consider a two-member household with members $i \in \{A, B\}$. Each member derives utility from private consumption goods $q^i \geq 0$, leisure $l_i \geq 0$, and a private asset position $\alpha_i \geq 0$ that represents financial autonomy. The household jointly consumes a public good $K \geq 0$ and a composite good $C = C^A + C^B + C^H \geq 0$, where C^A and C^B denote the shares consumed privately by each member and C^H denotes the jointly consumed share. Consumer durables and productive capital are examples of composite goods that generate utility for both members.

Each member is endowed with time T and faces a wage rate w^i, yielding labor income $w^i T$. Individual non-labor income is y^i, and the household receives joint non-labor income y^O. Total budget available to the household is:

$$Y = \sum_{i \in \{A,B\}} \left(y^i + w^i (T - l_i) \right) + y^O \tag{4}$$

Member i's preferences are represented by a utility function $U^i(q^i, l_i, \alpha_i, C, K)$, which is strictly increasing, twice continuously differentiable, and strictly quasi-concave in all arguments. We impose two assumptions that are relevant to the program context.

Assumption 1. *The household treats the loan $\mathcal{L}$ as an increase in joint income in the short term: $y'^O \mapsto y^O + \mathcal{L}$.*

Assumption 2. *Each member has a strict preference for leisure and financial autonomy: $\partial U^i / \partial l_i > 0$ and $\partial U^i / \partial \alpha_i > 0$.*

Assumption 1 reflects the context that farm production is organized at the household level and the program design where two household members co-signed the loan. Therefore, a loan for farming is potentially treated as a joint resource. Assumption 2 states that both leisure and financial autonomy are strictly valued.

11.2 Household Optimization

Under Pareto optimality, a household's allocation is characterized as the solution to a weighted utility maximization. Let $\mu \in (0, 1)$ denote member A's Pareto weight, making $1 - \mu$ as member B's weight. The household solves:

$$\max_{\{q^i, l_i, \alpha_i, C, K\}} \quad \mu\, U^A(q^A, l_A, \alpha_A, C, K) + (1 - \mu)\, U^B(q^B, l_B, \alpha_B, C, K) \tag{5}$$

subject to the household budget constraint:

$$\sum_i \left(p_q q^i + p_\alpha \alpha_i + w^i l_i\right) + p_C C + p_K K \leq Y \tag{6}$$

and the non-negativity constraints $q^i, l_i, \alpha_i, C, K \geq 0$ for all i.

The Pareto weight μ is not fixed but depends on the distribution of resources within the household. Following Browning and Chiappori (1998) and Bobonis (2009), we allow μ to be a function of the distribution factors. These are variables that affect bargaining power without entering preferences directly: $\mu = \mu(y^A, y^B, y^O)$ with $\partial\mu/\partial y^i \neq 0$ and $\partial\mu/\partial y^O \neq 0$. Under this formulation, the loan affects intrahousehold allocations through two distinct channels: it relaxes the household budget constraint (an income effect common to both members) and it shifts the Pareto weight (a distributional effect that reallocates resources between members).

11.3 Two-Stage Budgeting

The optimization problem includes a two-stage budgeting that distinguishes the loan's distinct effects on household-level and individual-level consumption.

Stage 1. Given total household resources Y and the Pareto weight μ, the household chooses expenditure on the public and composite goods. Let $C^*(Y, \mu)$ and $K^*(Y, \mu)$ denote the Stage 1 optima. By Stylized Fact 1, $\partial C^*/\partial Y > 0$, the loan raises composite good consumption for the household as a whole. If members value C and K, this channel generates a welfare gain for all members irrespective of how the loan affects bargaining.

Stage 2. Conditional on C^* and K^*, each member i chooses private consumption to maximize utility subject to an individual residual income constraint. Member i's residual income is:

$$\rho^i = y^i + w^i T - s^i \tag{7}$$

where $s^i \geq 0$ is member i's contribution to Stage 1 household expenditure, determined by an intrahousehold sharing rule $s^i = \phi^i(\mu, Y)$ with $\sum_i s^i = p_C C^* + p_K K^* - y^O$. Member i solves:

$$\max_{\{q^i, l_i, \alpha_i\}} \quad U^i(q^i, l_i, \alpha_i, C^*, K^*) \quad \text{s.t.} \quad p_q q^i + p_\alpha \alpha_i + w^i l_i \leq \rho^i \tag{8}$$

with C^* and K^* taken as given. The solution to (5) yields member i's indirect utility $V^i(\rho^i, C^*, K^*)$, which is strictly increasing in residual income, composite and public goods. The total effect of the loan on member i's welfare decomposes as:

$$\frac{dV^i}{d\mathcal{L}} = \underbrace{\frac{\partial V^i}{\partial C^*} \cdot \frac{\partial C^*}{\partial \mathcal{L}}}_{\geq 0} + \underbrace{\frac{\partial V^i}{\partial K^*} \cdot \frac{\partial K^*}{\partial \mathcal{L}}}_{\geq 0} + \underbrace{\frac{\partial V^i}{\partial \rho^i} \cdot \frac{d\rho^i}{d\mathcal{L}}}_{\gtrless 0} \tag{9}$$

The first two terms are weakly positive as both composite and public goods are unlikely increase individual members' disutility. The sign of the third term depends on how the loan affects member i's residual income.

11.4 The Effect of the Loan on Residual Income

The loan affects individual residual income in the short term in two ways. One is a bargaining channel, where if the loan shifts the bargaining rule in member i's favor, their required contribution s^i falls and ρ^i rises. The other is the resource expansion channel. As total household resources y^O grow, the total individual contributions required to finance joint expenditure fall, relaxing both members' private budget constraints. We characterize the welfare implications of the loan on individuals by considering three cases.

Case (i): $d\rho^i/d\mathcal{L} > 0$. The loan increases member i's residual income, either because it strengthens their bargaining position or because the income expansion channel reduces the burden of household expenditure contributions. All three terms in (6) are strictly positive. The loan is unambiguously welfare-improving for member i. By Assumption 3, the expanded residual income allows member i to increase leisure and financial autonomy above their pre-loan optima, generating welfare gains on all private consumption margins.

Case (ii): $d\rho^i/d\mathcal{L} = 0$. The loan leaves member i's residual income unchanged, as the bargaining and income expansion effects on s^i offset one another or neither operates. The welfare effect in (6) is positive as the first two terms are positive and third term equals 0. Member i benefits from the household-level consumption gains but experiences no change in private consumption, leisure, or financial autonomy. This case is consistent with a household that allocates the loan entirely to joint expenditure without altering the internal distribution of private resources.

Case (iii): $d\rho^i/d\mathcal{L} < 0$. The loan reduces member i's residual income, as the shift in bargaining power or the income expansion channel increases their required contribution s^i. The net welfare effect is now ambiguous. When the residual income loss is small relative to the gains in jointly consumed goods, $\Delta V^i > 0$ and the loan remains welfare-improving. When the loss is large enough to dominate, $\Delta V^i < 0$. The scenario in case (iii), where a positive household income shock leaves an individual member worse off, cannot be detected in a unitary model, further motivating the collective framework.

Appendix B: Outcome Variables

Household outcomes

Our primary household-level outcomes are binary variables of assets owned by the household and the World Food Programme's Food Consumption Score that measures diet

diversity. We analyze secondary outcomes of agricultural production to interpret impacts on household members' time use. These are area cultivated (in hectares), real income from farming, and household expenditure on inputs such as irrigation, seeds, fertilizers, labor, and land and machinery rental.

Individual outcomes

The individual level outcomes are categorized under financial inclusion, time use and involvement in decision making.

Financial inclusion: We expect if the loan expanded resources available to household members, it would be observable in their bank account activity and expression of financial autonomy through the use of ATM cards and mobile banking. We asked respondents whether they owned an individual bank account, used the debit card linked with this account on their own or shared it with their spouse and/or other family members. We also asked them to report how frequently they deposited/ withdrew funds or received salary or transfer payments in to their bank account. Finally, we asked if they had knowledge of and ever used mobile apps for banking. These binary indicators take the value 1 when individual responded yes, and 0 otherwise. Schaner (2016) found that debit cards are perceived to lower transaction costs of resource appropriation from individual bank accounts, and women reduced their account usage when provided a debit card. We investigate whether members are sharing their debit card or using it autonomously to infer their financial autonomy and control over resources. We measure their knowledge and use of mobile app-based banking because it enhances privacy and lowers the transaction costs of physically traveling to a bank branch or ATM for deposit or withdrawal. These apps are growing in popularity through low-cost internet plans and mobile phones, and steady promotion of digital transactions by the Indian government and the Reserve Bank of India (CGAP, 2019).

Time use: Hidrobo et al. (2016); Fink et al. (2020); El-Enbaby et al. (2025) show that loans and transfers to farming households impact the time use of members.The loan may affect household members' labor allocation to production activities either through expansion and intensification of input use or it may reduce demand for their labor through tools that allow members to work more productively. We asked household members to report the time they spent in the past day and during a busy day in the last agricultural season ("peak time") on various production activities such as farming, horticulture, livestock management, poultry, fishpond, and non-farm economic activities such as business, self-employment and wage employment. We also investigated non-production activities such as household purchases, domestic work and personal activities. Questions on time use were added to a module on participation in the baseline survey (that was in turn

adapted from the PRO-WEAI module and described below).[22] Time allocations under these activities do not add upto 24 because respondents were asked the time they spent on activities in Table B.1 conditional on them reporting they participated in that activity.

Decision-making: We use self-reported survey measures of decision making consistent with the literature measuring women's intrahousehold bargaining power (Allendorf, 2007; Connelly et al., 2010; Swaminathan et al., 2012; Field et al., 2021; Annan et al., 2021; Kosec et al., 2022; Heckert et al., 2023; Doss et al., 2022; Quisumbing et al., 2023a,b). We asked questions about members' involvement in the last 12 months in decisions around production and non-production activities such as farming, livestock and poultry, non-agricultural work and consumption (see Table B.1 for description). These questions were adopted from the project-level Women's Empowerment in Agriculture Index (pro-WEAI). We measure "involvement" as a combination of solo and joint decision making with other household members as we did not collect individual's preferences on which type of decisions they wanted to make.[23] We also test separately women's decision making jointly with their spouse.

We asked members to report the shares of their individual income from different production activities but discovered later that the question was administered incorrectly because the responses did not add up to 100. We do not test women's uptake of credit in response to the program as the loan was co-signed by two household members and not specifically targeted to any one household member.

[22] In the endline, we included the same activities from the baseline survey and added consumption decisions such as large and daily household purchase decisions, private consumption and personal activities (which includes leisure). Following the structure of the baseline survey, this module required a skip to the next activity if respondents said they did not participate. So respondents were not asked their time allocation if they said did not participate in that activity.

[23] Kochar et al. (2022) highlights that some decisions, such as those concerning food preparation, align with gender stereotypes about women's role in the household, and that delegating these tasks to women may reflect their lower agency.

Table B.1: Variable description of activities

Variable	Description
Farming	- Farming and processing of the harvest: grains that are grown primarily for food consumption (rice, maize, wheat, millet) - Horticultural (gardens) or high value crop farming and processing of the harvest
Livestock & Poultry	- Raising cattle, buffaloes, sheep, goats, pigs and processing of their milk and/or meat; raising chickens, ducks, turkeys and processing their eggs and/or meat
Non-agriculture	- Small business, self-employment, buy-and-sell - Wage and salary employment
Household purchases	- Large, occasional household purchases (bicycles, land, transport vehicles) - Routine household purchases (food for daily consumption or other household needs)
Domestic work	Cooking, cleaning, fetching firewood or water, and caring for children or others
Personal activity	Eating, leisure such as visiting neighbors, watching TV, listening to the radio, seeing movies, or doing sports

Notes: Variable generation for analyzing time use and decision making from individual activities asked in the survey.

Online Appendix C. Figures

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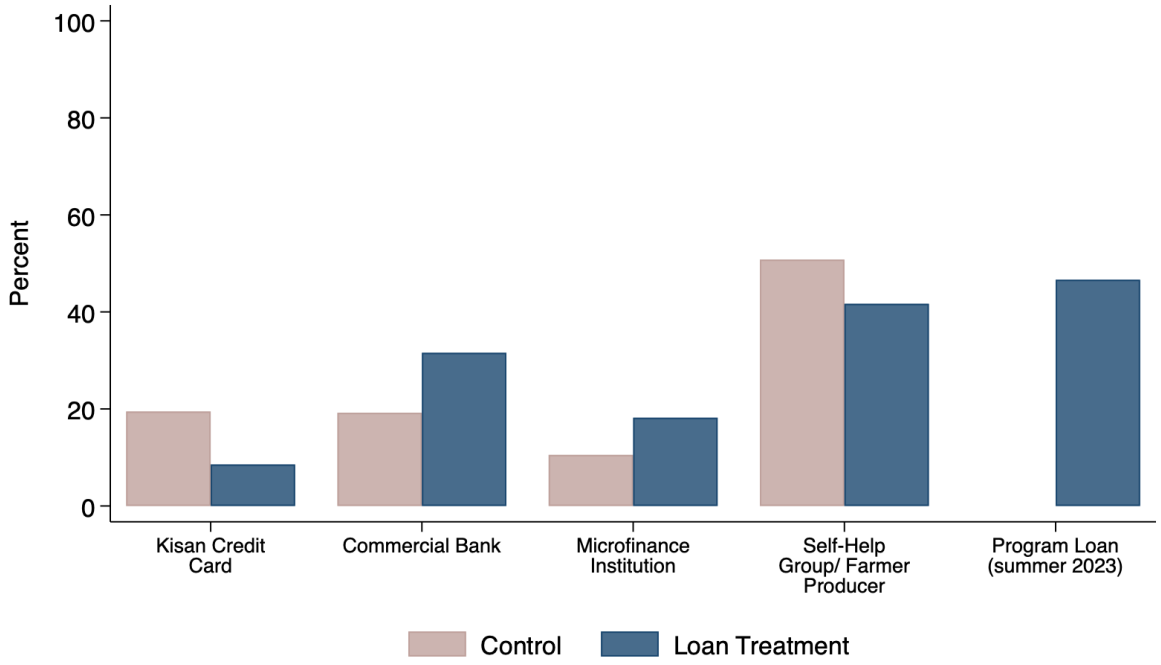


Figure C.1: Sources of formal loan by intervention arm

Notes: The figure plots the distribution of uptake of formal sources of credit for the summer planting season in 2023 reported by households in the endline survey.

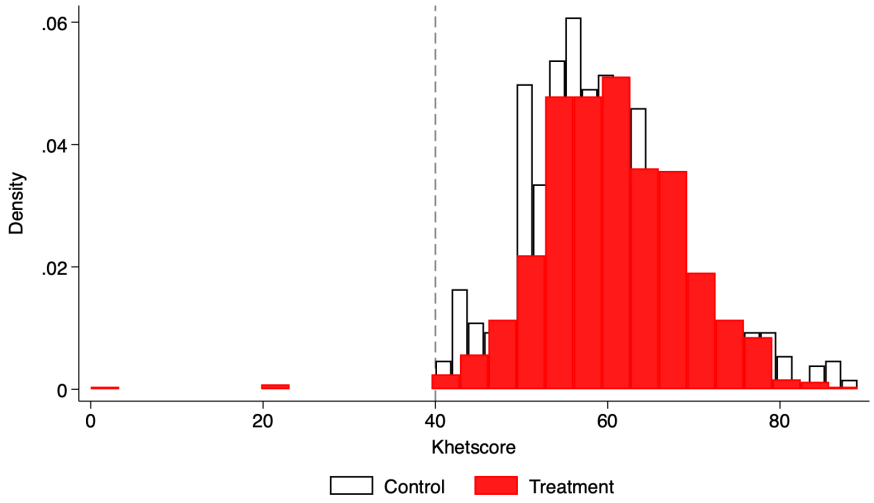


Figure C.2: Distribution of sample by credit score

Notes: The figure plots the distribution of credit score of households in the control (black outline) and treatment arm (solid red). The score is a weighted index computed by DER and the loan eligibility threshold was 40.

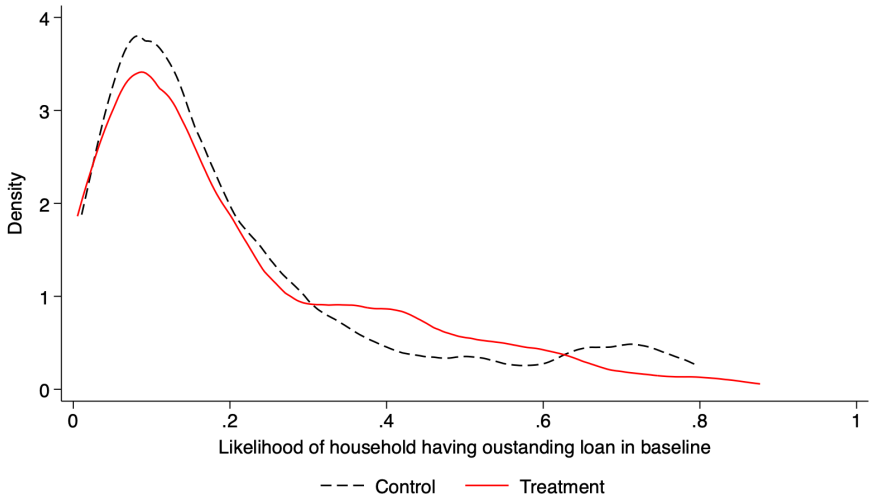


Figure C.3: Distribution of propensity to borrow by intervention arm

Notes: The figure plots the propensity score distributions of households receiving any loan at baseline in the control (dashed line) and treatment arm (solid line). The score is estimated using a logit model where the outcome is a binary variable of whether the household had an outstanding loan and the explanatory variables are household characteristics from the baseline survey.

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Table D.1: Cost of the loan program and net returns

Fixed costs (per season)		
Project management	8733.33	
Staff travel	1071.33	
Field office cost	4000.00	
App/portal maintenance	5158.67	
Communication material	178.67	
Total fixed cost per season	19142.00	
Total fixed cost per farmer	30.58	
	Maharashtra	Odisha
Variable costs (per farmer)		
Khetscore report	5.5	5.5
DER field staff	10.56	10.56
Staff transport, allowance	2.079	2.156
Printing, stationary	0.352	0.363
Total Variable cost per farmer	18.491	18.579
Interest 14% of loan	72.38	63.14
Number of beneficiaries	160	193
Defaulters	0	3
Total costs= (fixed + variable)	7851.36	9487.69
Total revenue	11580.80	11996.60
Net return	3729.44	2508.91
Net return per farmer	23.31	13.00

Notes: The cost of the program (in USD) is compiled using data shared by DER. Total cost per farmer is estimated using total farmers who received the loan product across the entire intervention, which included two treatment arms. This paper only analyzes the loan treatment arm while Kramer et al. (2025a) documents the additional loan and insurance treatment.

Table D.2: Attrition by treatment assignment and baseline characteristics

	Sample that attrited	
	(1)	(2)
Treatment: Loan	-0.038	-0.033
	(0.036)	(0.037)
Respondents' age		0.001
		(0.002)
Respondent is female		0.061**
		(0.025)
Can read and write		0.027
		(0.029)
Completed at least primary school		-0.008
		(0.027)
Completed at least secondary school		-0.013
		(0.025)
Respondent belongs to SC/ST category		-0.023
		(0.022)
Respondent is married		0.021
		(0.031)
Annual Household Income(<90,000)		0.024
		(0.029)
Annual Household Income (> 120,000)		0.024
		(0.024)
Owns land		-0.004
		(0.024)
Respondent has bank account		-0.044
		(0.044)
Borrowed money last 12 months		0.018
		(0.024)
Have outstanding loan		-0.059*
		(0.033)
Constant	0.152***	0.134
	(0.023)	(0.083)
Observations	1682	1682
Non-attrited (N)	1431	1431
Block dummies	Yes	Yes
Mean dep. variable	0.149	

Notes: The table reports reduced form estimation of whether attrition of the sample was significantly correlated with treatment assignment (column 1) or ex-ante characteristics of the household measured from the baseline survey (column 2). The binary dependent variable distinguishes the sample that was interviewed at baseline and not in the endline survey. Both models include block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. The average sample that attrited in the control group is reported in the last row of column 1. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table D.3: Administrative data on loan take-up

	Summer 2023	
	Control	Loan Treatment
	(1)	(2)
DER generated Credit score	58.9	60.5
Applied for loan	0	58.5%
Received loan	0	47%
Loan amount (INR)	0	41, 505
Default		0.85%
Observations	681	748

Notes: The table reports the disbursal of the program's loan product to households in the control and treatment arm. Column 1 verifies the validity of the experiment design as no control group households were given the loan. This data is extracted from the loan records of the implementing partner DER.

Table D.4: Balance test of outcome variables at baseline

		(1)		(2)		(3)	F-test for balance	(2)-(3)
		All		Loan		Control	across all groups	Pairwise t-test
	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	F-stat/P-value	Mean difference
Income	1319	1071.018	701	1090.449	618	1048.979	0.01	41.47
	80	(-208.991)	40	(-351.899)	40	(-206.698)	(0.919)	
Income main crop	928	1202.374	487	1057.025	441	1362.884	1.732	-305.859
	78	(-120.477)	39	(-167.085)	39	(-163.539)	(0.192)	
Harvest (quintiles)	1429	35.424	748	36.93	681	33.769	0.902	3.162
	80	(-1.724)	40	(-2.564)	40	(-2.155)	(0.345)	
Harvest main crop (quintiles)	850	30.595	415	29.075	435	32.045	0.884	-2.97
	68	(-1.586)	35	(-2.178)	33	(-2.319)	(0.351)	
Yield (Harvest/ Area)	1361	14.394	722	15.336	639	13.33	0.895	2.006
	80	(-1.078)	40	(-1.46)	40	(-1.556)	(0.347)	
Yield (Harvest/ Area) main crop	850	10.01	415	10.176	435	9.852	0.041	0.325
	68	(-0.807)	35	(-1.13)	33	(-1.163)	(0.841)	
Expenditure male labor	1067	27.125	563	37.863	504	15.129	7.245***	22.734***
	80	(-4.778)	40	(-7.711)	40	(-3.568)	(0.009)	
Expenditure female labor	1067	6.196	563	9.526	504	2.475	8.226***	7.051***
	80	(-1.379)	40	(-2.371)	40	(-0.704)	(0.005)	
Expenditure male labor	1429	53.858	748	58.622	681	48.625	3.205*	9.998*
main crop	80	(-2.903)	40	(-3.585)	40	(-4.326)	(0.077)	
Expenditure female labor	1429	48.85	748	46.412	681	51.528	0.18	-5.115
main crop	80	(-6.03)	40	(-8.037)	40	(-9.09)	(0.673)	
Irrigation cost	1429	20.133	748	19.765	681	20.536	0.033	-0.771
	80	(-2.163)	40	(-3.326)	40	(-2.722)	(0.857)	
Irrigation cost main crop	1429	17.371	748	16.541	681	18.283	0.274	-1.741
	80	(-1.693)	40	(-2.588)	40	(-2.124)	(0.602)	
Seed cost	1429	42.759	748	36.992	681	49.094	1.148	-12.101
	80	(-5.666)	40	(-6.951)	40	(-8.985)	(0.287)	
Seed cost main crop	1429	40.888	748	35.181	681	47.157	1.264	-11.976
	80	(-5.345)	40	(-6.442)	40	(-8.562)	(0.264)	

(Contd.)

		(1)		(2)		(3)	F-test for balance	(2)-(3)
		All		Loan		Control	across all groups	Pairwise t-test
	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	F-stat/P-value	Mean difference
	80	(-0.034)	40	(-0.051)	40	(-0.042)	(0.221)	
Fertilizer cost	1429	101.535	748	92.54	681	111.415	0.482	-18.875
	80	(-13.58)	40	(-17.569)	40	(-20.965)	(0.49)	
Fertilizer cost main crop	1429	80.353	748	65.773	681	96.368	2.757	-30.595
	80	(-9.085)	40	(-7.938)	40	(-16.752)	(0.101)	
Rental cost machinery	1429	55.082	748	54.524	681	55.695	0.01	-1.171
	80	(-5.94)	40	(-8.3)	40	(-8.613)	(0.922)	
Rental cost machinery main crop	1429	47.103	748	43.396	681	51.175	0.948	-7.78
	80	(-3.896)	40	(-3.725)	40	(-7.124)	(0.333)	
Land rent	1429	6.076	748	7.41	681	4.611	1.281	2.799
	80	(-1.265)	40	(-1.886)	40	(-1.621)	(0.261)	
Land rent main crop	1429	6.076	748	7.41	681	4.611	1.281	2.799
	80	(-1.265)	40	(-1.886)	40	(-1.621)	(0.261)	
Days own labor (num)	1429	27.769	748	28.263	681	27.226	0.059	1.037
	80	(-2.124)	40	(-2.816)	40	(-3.232)	(0.808)	
Days own labor main crop	1429	25.731	748	26.476	681	24.913	0.172	1.563
	80	(-1.899)	40	(-2.736)	40	(-2.616)	(0.679)	
Days male HH member's labor (num)	1429	17.624	748	16.311	681	19.066	1.231	-2.755
	80	(-1.256)	40	(-1.666)	40	(-1.86)	(0.271)	
Days male HH member's	1429	16.283	748	15.131	681	17.548	1.244	-2.417
labor main crop	80	(-1.1)	40	(-1.535)	40	(-1.548)	(0.268)	
Days female HH member's labor (num)	1429	14.877	748	14.211	681	15.608	0.204	-1.397
	80	(-1.549)	40	(-2.149)	40	(-2.251)	(0.653)	
Days female HH member's	1429	14.076	748	13.357	681	14.865	0.272	-1.508
labor main crop	80	(-1.446)	40	(-1.965)	40	(-2.147)	(0.604)	
Male wage labor (mean) kharif	1287	2.781	675	2.762	612	2.801	0.056	-0.039
	69	(-0.083)	35	(-0.119)	34	(-0.117)	(0.813)	
Female wage labor (mean) kharif	1287	1.986	675	1.999	612	1.972	0.03	0.028
	69	(-0.081)	35	(-0.127)	34	(-0.099)	(0.863)	

Notes: The table reports outcomes measured at baseline for the main crop cultivated during summer of 2022. All expenditure variables are in INR and deflated to real values. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.5: LATE and ITT: Time use (hours during peak time conditional on participation)

	Personal activity (1)	Non-agriculture (2)	Farming (3)	Livestock & Poultry (4)	Consumption (5)	Domestic work (6)
Panel A: Women						
LATE: Received loan product (γ_F)	0.939**	-0.609	-0.299	-0.204	-0.002	-0.439
	(0.409)	(0.996)	(1.749)	(0.578)	(0.390)	(0.471)
	{0.178}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}
ITT: Village assigned to treatment (β_F)	0.433***	-0.302	-0.123	-0.093	-0.001	-0.204
	(0.146)	(0.486)	(0.720)	(0.265)	(0.178)	(0.224)
	{0.027}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}
First-stage F	31.143	36.932	60.189	56.260	31.391	35.821
Observations	443	283	615	179	439	453
Control group mean	1.676	4.476	4.932	1.985	1.721	3.188
Panel B: Men						
LATE: Received loan product (γ_M)	0.256	0.584	1.211*	0.465	-0.331	0.964**
	(0.197)	(0.621)	(0.723)	(0.667)	(0.379)	(0.457)
	{0.417}	{0.446}	{0.325}	{0.483}	{0.446}	{0.312}
ITT: Village assigned to treatment (β_M)	0.123	0.337	0.578*	0.198	-0.156	0.448**
	(0.091)	(0.349)	(0.331)	(0.276)	(0.176)	(0.191)
	{0.342}	{0.436}	{0.270}	{0.466}	{0.436}	{0.157}
First-stage F	114.137	148.102	175.615	39.704	127.039	84.039
Observations	528	418	989	273	502	311
Control group mean	1.602	3.995	5.456	2.271	1.697	1.611
LATE difference ($\gamma_F - \gamma_M$)	0.684	-1.193	-1.510	-0.669	0.329	-1.403
p-value of test ($\gamma_F = \gamma_M$)	0.132	0.309	0.425	0.448	0.545	0.033
ITT difference ($\beta_F - \beta_M$)	0.310	-0.639	-0.700	-0.291	0.155	-0.652
p-value of test ($\beta_F = \beta_M$)	0.072	0.285	0.377	0.447	0.536	0.027
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Interview month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table repeats Table 7 for respondents who participated in the activity. The dependent variables are listed as column titles (see Table B.1 for description). Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table D.6: LATE and ITT: Time use (hours yesterday conditional on participation)

	Personal activity (1)	Non-agriculture (2)	Farming (3)	Livestock & Poultry (4)	Consumption (5)	Domestic work (6)
Panel A: Women						
LATE: Received	0.856**	2.092*	-1.255	0.103	-0.051	-0.945*
loan product (γ_F)	(0.372)	(1.152)	(1.674)	(0.573)	(0.529)	(0.544)
	{0.177}	{0.177}	{0.521}	{0.859}	{0.859}	{0.177}
ITT: Village	0.392***	1.057*	-0.502	0.045	-0.023	-0.450*
assigned to	(0.141)	(0.573)	(0.682)	(0.248)	(0.240)	(0.239)
treatment (β_F)	{0.046}	{0.134}	{0.535}	{0.859}	{0.859}	{0.134}
First-stage F	34.106	39.781	61.696	24.254	34.849	41.673
Observations	408	244	566	178	400	416
Control group mean	1.788	3.901	3.748	2.199	3.841	3.528
Panel B: Men						
LATE: Received	0.004	0.926	0.906	0.515	0.155	-0.093
loan product (γ_M)	(0.228)	(0.577)	(0.655)	(0.586)	(0.799)	(0.419)
	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}
ITT: Village	0.002	0.532	0.432	0.211	0.072	-0.043
assigned to	(0.113)	(0.319)	(0.310)	(0.238)	(0.370)	(0.195)
treatment (β_M)	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}	{1.000}
First-stage F	126.576	145.716	171.618	46.925	118.132	82.892
Observations	536	410	958	270	493	321
Control group mean	2.016	4.120	3.921	2.426	3.946	1.866
LATE difference ($\gamma_F - \gamma_M$)	0.852	1.165	-2.162	-0.412	-0.206	-0.851
p-value of test ($\gamma_F = \gamma_M$)	0.051	0.366	0.229	0.615	0.830	0.215
ITT difference ($\beta_F - \beta_M$)	0.389	0.525	-0.934	-0.166	-0.095	-0.407
p-value of test ($\beta_F = \beta_M$)	0.031	0.424	0.212	0.629	0.830	0.187
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Interview month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table repeats Table 8 for respondents who participated in the activity. The dependent variables are listed as column titles (see Table B.1 for description). Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table D.7: ITT: Loan intervention and panel respondent's report of labor utilization

	All crops			Main crops		
	Own labor	Male household member	Female household member	Own labor	Male household member	Female household member
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment: Loan	-0.448	0.426	0.901	-3.920	-2.110	0.188
	(4.652)	(3.419)	(3.107)	(4.625)	(3.087)	(3.000)
Sharpened q-values	1	1	1	1	1	1
Observations	1429	1429	1429	1429	1429	1429
Mean dependent variable (Baseline)	27.769	17.624	14.877	25.731	16.283	14.076
Covariates included	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports the ITT effect of the loan intervention on number of days of labor utilization during the summer agricultural season. Columns 1-3 estimate for all crops and columns 4-6 are restricted to the main crop (rice in Odisha and cotton and soybean in Maharashtra). The dependent variables are continuous and listed as column titles. The model includes outcome variable and covariates measured at baseline as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.8: ITT: Loan intervention and agricultural expansion

	Real income from farming	Harvest	Yield (Harvest/ area)
	(1)	(2)	(3)
Panel A: All crops			
Treatment: Loan	679.596	21.208	1.912
	(678.788)	(16.477)	(3.287)
Sharpened q-values	0.316	0.254	0.508
Observations	1265	1429	1361
Mean dependent variable (Baseline)	1071.018	35.424	14.394
Covariates included	Yes	Yes	Yes
Block FE	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes
Panel B: Main crops (paddy/ cotton/ soybean)			
Treatment: Loan	-93.875	0.253	-1.270
	(114.717)	(2.464)	(1.852)
Sharpened q-values	1	1	1
Observations	847	802	802
Mean dependent variable (Baseline)	1202.374	30.595	10.010
Covariates included	Yes	Yes	Yes
Block FE	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes

Notes: The table reports the ITT effect of the loan intervention on agricultural expansion, output and productivity during the summer agricultural season. Panel A includes results for all crops and panel B incudes only for the main crop (rice in Odisha and cotton and soybean in Maharashtra). The dependent variables are continuous and listed as column titles. The model includes outcome variable and covariates measured at baseline as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.9: ITT: Loan intervention and wages

	Aggregate real wages (all crops)		Total real expenditure (main crops)	
	Male	Female	Male	Female
	(1)	(2)	(3)	(4)
Treatment: Loan	-0.034	0.004	-12.561	-6.715
	(0.065)	(0.076)	(15.768)	(12.137)
Sharpened q-values	1	1	1	1
Observations	1161	1160	1429	1429
Mean dependent variable (Baseline)	2.781	1.986	53.858	48.850
Covariates included	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes

Notes: The table reports the ITT effect of the loan intervention on the wage rate aggregated over all crops (columns 1-2) and total real expenditure on labor for the main crop by a household during the summer agricultural season. The main crop constitutes rice in Odisha and cotton and soybean in Maharashtra. The dependent variables are continuous and listed as column titles. The model includes outcome variable and covariates measured at baseline as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.10: ITT: Loan intervention and expenditure on inputs

	Total expenditure (real terms)				
	Irrigation	Seeds	Fertilizer	Machines	Rent
	(1)	(2)	(3)	(4)	(5)
Panel A: All crops					
Treatment: Loan	5.326	5.898	33.777	-7.962	3.480
	(3.383)	(5.241)	(35.955)	(11.915)	(6.724)
Sharpened q-values	1	1	1	1	1
Observations	1429	1429	1429	1429	1429
Mean dependent variable (Baseline)	20.133	42.759	101.535	55.082	6.076
Covariates included	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes	Yes
Panel B: Main crops (paddy/ cotton/ soybean)					
Treatment: Loan	3.443	4.064	-9.244	-16.153	3.458
	(3.384)	(4.735)	(14.561)	(12.033)	(6.469)
Sharpened q-values	1	1	1	1	1
Observations	1429	1429	1429	1429	1429
Mean dependent variable (Baseline)	17.371	40.888	80.353	47.103	6.076
Covariates included	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes	Yes

Notes: The table reports the ITT effect of the loan intervention on purchase of inputs for cultivation of the main crop during during the summer agricultural season. Panel A includes results for all crops and panel B incudes only for the main crop (rice in Odisha and cotton and soybean in Maharashtra). The dependent variables are continuous and listed as column titles. The model includes outcome variable and covariates measured at baseline as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.11: LATE: Use of formal loans

	Seeds	Pesticides	Labor	Rent machinery	Repayment of another loan
	(1)	(2)	(3)	(4)	(5)
Received loan product	-0.041	-0.196	-0.016	-0.193	-0.318**
	(0.112)	(0.124)	(0.184)	(0.152)	(0.155)
Constant	0.465**	1.419***	1.025***	1.440***	0.175
	(0.235)	(0.354)	(0.326)	(0.298)	(0.299)
Sharpened q value	0.556	0.296	0.596	0.377	0.258
Observations	608	608	608	608	608
Control group mean	0.896	0.906	0.625	0.802	0.432
Covariates included	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of use of formal loans on categories listed as column titles. Each dependent variable is binary. The information was extracted from the endline survey where households were asked how they spent their formal loan(s). The analysis uses treatment assignment as an exogenous instrument for a household's receipt of the loan product. The model includes covariates measured at baseline as regressors and block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.12: LATE and ITT by state, women: Individual bank account and debit card use

	Own bank account (1)	Deposited money (2)	Withdrew money (3)	Received money (4)	Use debit card alone (5)	Share debit card (6)
Panel A: Odisha						
LATE: Received	0.115	0.585***	0.591***	0.264*	0.843***	-0.187*
loan product (γ_F)	(0.089)	(0.153)	(0.149)	(0.137)	(0.178)	(0.109)
	{0.100}	{0.001}	{0.001}	{0.048}	{0.001}	{0.060}
ITT: Village	0.052	0.265***	0.268***	0.119**	0.381***	-0.085*
assigned to	(0.038)	(0.061)	(0.061)	(0.059)	(0.059)	(0.049)
treatment (β_F)	{0.079}	{0.001}	{0.001}	{0.038}	{0.001}	{0.060}
First-stage F	55.088	55.088	55.088	55.088	55.088	55.088
Observations	736	736	736	736	736	736
Control group mean	0.866	0.262	0.331	0.240	0.178	0.379
Panel B: Maharashtra						
LATE: Received	-0.233	0.189*	0.334***	0.272***	0.516***	-0.089
loan product (γ_F)	(0.214)	(0.097)	(0.109)	(0.095)	(0.172)	(0.134)
	{0.129}	{0.047}	{0.014}	{0.014}	{0.014}	{0.206}
ITT: Village	-0.132	0.107*	0.189***	0.154***	0.292**	-0.050
assigned to	(0.112)	(0.059)	(0.060)	(0.056)	(0.114)	(0.074)
treatment (β_F)	{0.131}	{0.062}	{0.019}	{0.024}	{0.025}	{0.200}
First-stage F	66.142	66.142	66.142	66.142	66.142	66.142
Observations	390	390	390	390	390	390
Control group mean	0.842	0.557	0.566	0.561	0.176	0.195
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table repeats Table 5 for women only, estimated separately by state. Panel A restricts the sample to women in Odisha and Panel B to women in Maharashtra. Sharpened q-values (curly braces) are computed separately within each state panel, so a family consists of the outcomes within one state panel, and LATE and ITT estimates are treated as separate families. Standard errors are clustered at the village level. All other details follow Table 5. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table D.13: LATE and ITT by state, women: mobile phone ownership and and digital financial services

	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)
Panel A: Odisha			
LATE: Received	0.347**	1.099***	0.923***
loan product (γ_F)	(0.160)	(0.180)	(0.153)
	{0.013}	{0.001}	{0.001}
ITT: Village	0.167**	0.497***	0.418***
assigned to	(0.077)	(0.054)	(0.055)
treatment (β_F)	{0.013}	{0.001}	{0.001}
First-stage F	47.372	55.088	55.088
Observations	484	736	736
Control group mean	0.302	0.220	0.187
Panel B: Maharashtra			
LATE: Received	-0.218	0.553***	0.612***
loan product (γ_F)	(0.189)	(0.165)	(0.165)
	{0.094}	{0.003}	{0.003}
ITT: Village	-0.114	0.313***	0.347***
assigned to	(0.095)	(0.111)	(0.113)
treatment (β_F)	{0.088}	{0.012}	{0.012}
First-stage F	53.003	66.142	66.142
Observations	310	390	390
Control group mean	0.299	0.217	0.181
Baseline controls	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes

Notes: The table repeats Table 6 for women only, estimated separately by state. Panel A restricts the sample to women in Odisha and Panel B to women in Maharashtra. Sample sizes differ as non-response to the mobile ownership questions are marked missing. Sharpened q-values (curly braces) are computed separately within each state panel, so a family consists of the outcomes within one state panel, and LATE and ITT estimates are treated as separate families. Standard errors are clustered at the village level. All other details follow Table 6. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table D.14: LATE: Transactions from bank account

	Withdraw 2-3 times/ month (1)	Deposit 2-3 times/ month (2)	Receive money 2-3 times/ month (3)
Panel A: Women			
Received loan product (γ_F)	0.442***	0.399***	0.244**
	(0.101)	(0.106)	(0.101)
Constant	-0.136	-0.053	-0.114
	(0.186)	(0.174)	(0.168)
Sharpened q-values	0.001	0.001	0.006
Observations	1126	1126	1126
Control group mean	0.421	0.374	0.362
Panel B: Men			
Received loan product (γ_M)	0.272**	0.347***	0.274**
	(0.114)	(0.121)	(0.123)
Constant	0.250	0.180	0.296
	(0.193)	(0.260)	(0.194)
Sharpened q-values	0.018	0.013	0.018
Observations	1364	1364	1364
Control group mean	0.400	0.303	0.217
Difference ($\gamma_F - \gamma_M$)	0.170	0.053	-0.030
p-value of test ($\gamma_F = \gamma_M$)	0.266	0.744	0.849
Baseline Controls	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes

Notes: The table reports LATE estimates of the loan product on household members' withdrawal, deposit and money receipt in their bank account. We asked household members their frequency of account deposits and withdrawals and found 44% of them deposited funds in their account 2-3 times a month and about 51% withdrew funds with the same frequency, while weekly deposit and withdrawal were less than 3%. Therefore, we analyze differences in account transactions with the frequency of 2-3 times a month. LATE estimates are from a 2SLS model using village assignment to treatment as an instrument for a household's receipt of the loan product. The dependent variables (binary) are listed as column titles. Panel A restricts the sample to women in the endline survey and Panel B to men. All regressions control for the household's demographic and economic characteristics from the baseline survey and include block fixed effects. The first-stage F is the F statistic for regressing village assignment on household's receipt of the loan product, conditional on the controls and fixed effects. Standard errors in all models are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q-values are reported in curly braces and a family consists of the three outcomes within a panel. The test of equality of coefficients between women and men assumes independence between the two samples. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table D.15: LATE: Heterogeneous effects in reports of time allocations (peak time) by individual account ownership

	Farming	Livestock & Poultry	Non-agriculture	Consumption	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan * Own bank account	-1.856	0.257	-0.829	-1.501***	-1.111	1.250*
	(1.555)	(0.562)	(0.773)	(0.481)	(1.365)	(0.653)
Received Loan	-0.739	-0.400	-1.451**	1.296***	0.567	-0.207
	(1.551)	(0.490)	(0.704)	(0.445)	(1.145)	(0.753)
Own bank account	1.321***	-0.267	0.349	-0.007	-0.244	-0.347
	(0.474)	(0.210)	(0.255)	(0.139)	(0.481)	(0.326)
Constant	0.245	0.461	-1.353	-0.760*	4.062***	1.765***
	(1.160)	(0.327)	(0.940)	(0.446)	(1.040)	(0.561)
Sharpened q-values	0.884	0.884	0.884	0.884	0.884	0.884
Observations	1119	1119	1119	1119	453	443
Control group mean	3.130	0.356	1.323	0.677	3.188	1.676
Panel B: Men						
Received loan * Own bank account	0.601	0.417	3.619**	0.693	2.834***	0.590
	(1.478)	(0.538)	(1.465)	(0.588)	(1.054)	(0.510)
Received Loan	-0.123	-0.641	-3.147**	-0.674	-1.232*	-0.233
	(1.124)	(0.467)	(1.507)	(0.580)	(0.705)	(0.487)
Own bank account	0.345	-0.119	-0.428	-0.678***	-0.337	-0.122
	(0.578)	(0.181)	(0.452)	(0.200)	(0.298)	(0.111)
Constant	1.482	-0.481	3.312***	2.194***	2.264**	1.782***
	(1.324)	(0.451)	(1.017)	(0.418)	(0.906)	(0.461)
Sharpened q-values	1	1	1	1	0.946	1
Observations	1353	1353	1353	1353	311	528
Control group mean	4.144	0.489	1.247	0.677	1.611	1.602
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table repeats Table 7. We interact loan receipt with women's individual account ownership, and then instrument loan receipt and its interaction with village assignment and its interaction with the same covariate. Panel A restricts the sample to women Panel B to men. Sharpened q-value of the interaction term are computed separately within each gender panel, so a family consists of the outcomes within one gender panel. Standard errors are clustered at the village level. Inclusion of controls and fixed effects follow Table 7. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table D.16: LATE: Heterogeneous effects in reports of time allocations (yesterday) by individual account ownership

	Farming	Livestock & Poultry	Non-agriculture	Consumption	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan * Own bank account	-0.774	0.342	0.753	-4.131***	1.872	0.998
	(1.080)	(0.691)	(0.775)	(1.323)	(1.265)	(0.765)
Received Loan	-1.573	-0.498	-1.663**	3.388***	-2.516**	-0.032
	(1.067)	(0.586)	(0.651)	(1.255)	(1.111)	(0.714)
Own bank account	0.956**	-0.363	0.105	0.167	-1.361***	-0.463
	(0.418)	(0.282)	(0.279)	(0.267)	(0.485)	(0.342)
Constant	-0.698	0.182	-1.362*	-0.272	4.813***	2.251***
	(0.859)	(0.324)	(0.789)	(1.292)	(0.924)	(0.585)
Sharpened q-values	0.493	0.493	0.493	0.493	0.511	0.493
Observations	1081	1081	1081	1081	416	408
Control group mean	2.244	0.402	0.953	1.326	3.528	1.788
Panel B: Men						
Received loan * Own bank account	-0.190	0.401	2.215*	0.149	1.945*	0.758
	(1.268)	(0.504)	(1.154)	(1.977)	(1.089)	(0.474)
Received Loan	0.465	-0.698	-1.591	0.137	-1.697**	-0.520
	(1.026)	(0.451)	(1.073)	(2.042)	(0.840)	(0.411)
Own bank account	0.505	-0.060	0.132	-1.800***	-0.136	-0.419***
	(0.452)	(0.165)	(0.246)	(0.451)	(0.286)	(0.155)
Constant	0.996	-0.582	2.011**	6.832***	2.418***	3.000***
	(0.960)	(0.555)	(0.900)	(1.323)	(0.774)	(0.706)
Sharpened q-values	0.852	0.852	0.961	0.961	0.513	0.513
Observations	1345	1345	1345	1345	321	536
Control group mean	2.842	0.521	1.272	1.443	1.866	2.016
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table repeats Table 8. We interact loan receipt with women's individual account ownership, and then instrument loan receipt and its interaction with village assignment and its interaction with the same covariate. Panel A restricts the sample to women Panel B to men. Sharpened q-value of the interaction term are computed separately within each gender panel, so a family consists of the outcomes within one gender panel. Standard errors are clustered at the village level. Inclusion of controls and fixed effects follow Table 8. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table D.17: LATE and ITT by state, women: Time use (peak agricultural season)

	Personal activity (1)	Non-agriculture (2)	Farming (3)	Livestock & Poultry (4)	Consumption (5)	Domestic work (6)
Panel A: Odisha						
LATE: Received	2.964*	0.138	-0.758	-0.428	0.813*	1.861
loan product (γ_F)	(1.489)	(0.349)	(1.704)	(0.269)	(0.447)	(2.123)
	{0.298}	{0.533}	{0.533}	{0.298}	{0.298}	{0.412}
ITT: Village	0.789***	0.061	-0.336	-0.190	0.360**	0.478
assigned to	(0.241)	(0.159)	(0.772)	(0.120)	(0.166)	(0.472)
treatment (β_F)	{0.019}	{0.542}	{0.542}	{0.196}	{0.099}	{0.325}
First-stage F	8.366	53.061	53.061	53.061	53.061	9.150
Observations	195	734	734	734	734	190
Control group mean	0.908	0.214	2.724	0.380	0.298	2.601
Panel B: Maharashtra						
LATE: Received	0.101	-4.106***	-3.892***	-0.159	-0.838***	-0.358
loan product (γ_F)	(0.178)	(0.668)	(0.963)	(0.132)	(0.197)	(0.438)
	{0.405}	{0.001}	{0.001}	{0.217}	{0.001}	{0.336}
ITT: Village	0.061	-2.486***	-2.357***	-0.096	-0.507***	-0.218
assigned to	(0.106)	(0.415)	(0.568)	(0.080)	(0.137)	(0.281)
treatment (β_F)	{0.398}	{0.001}	{0.001}	{0.218}	{0.001}	{0.362}
First-stage F	38.242	71.118	71.118	71.118	71.118	35.282
Observations	248	385	385	385	385	263
Control group mean	1.988	3.167	3.806	0.315	1.306	3.429
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Interview month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table repeats Table 7 for women only, estimated separately by state. See Table B.1 for description of column titles. Panel A restricts the sample to women in Odisha and Panel B to women in Maharashtra. Sharpened q-values (curly braces) are computed separately within each state panel, so a family consists of the outcomes within one state panel, and LATE and ITT estimates are treated as separate families. Standard errors are clustered at the village level. All other details follow Table 7. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table D.18: LATE and ITT by state, women: Time use (hours yesterday)

	Personal activity (1)	Non-agriculture (2)	Farming (3)	Livestock & Poultry (4)	Consumption (5)	Domestic work (6)
Panel A: Odisha						
LATE: Received loan product (γ_F)	1.899*	0.861	-1.114	-0.485	0.652	-0.286
	(1.022)	(0.512)	(1.385)	(0.317)	(0.394)	(1.445)
	{0.252}	{0.252}	{0.252}	{0.252}	{0.252}	{0.392}
ITT: Village assigned to treatment (β_F)	0.536***	0.384	-0.497	-0.217	0.291*	-0.088
	(0.190)	(0.256)	(0.642)	(0.142)	(0.151)	(0.454)
	{0.058}	{0.215}	{0.269}	{0.215}	{0.183}	{0.394}
First-stage F	9.673	55.411	55.411	55.411	55.411	11.814
Observations	199	735	735	735	735	191
Control group mean	1.235	0.334	1.955	0.416	0.269	3.290
Panel B: Maharashtra						
LATE: Received loan product (γ_F)	0.400	-2.859***	-3.409***	-0.156	-1.451*	-0.660
	(0.268)	(0.577)	(0.783)	(0.144)	(0.808)	(0.392)
	{0.133}	{0.001}	{0.001}	{0.181}	{0.114}	{0.114}
ITT: Village assigned to treatment (β_F)	0.244	-1.713***	-2.042***	-0.093	-0.869*	-0.403*
	(0.155)	(0.347)	(0.462)	(0.087)	(0.505)	(0.218)
	{0.111}	{0.001}	{0.001}	{0.164}	{0.104}	{0.104}
First-stage F	43.508	72.162	72.162	72.162	72.162	38.888
Observations	209	346	346	346	346	225
Control group mean	2.099	2.209	2.831	0.373	3.469	3.654
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Interview month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table repeats Table 8 for women only, estimated separately by state. See Table B.1 for description of column titles. Panel A restricts the sample to women in Odisha and Panel B to women in Maharashtra. Sharpened q-values (curly braces) are computed separately within each state panel, so a family consists of the outcomes within one state panel, and LATE and ITT estimates are treated as separate families. Standard errors are clustered at the village level. All other details follow Table 8. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table D.19: ITT by state: Loan intervention and panel respondent's report of labor utilization

	All crops			Main crops		
	Own labor	Male household member	Female household member	Own labor	Male household member	Female household member
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Odisha						
Treatment: Loan	-0.135	-1.044	0.209	-0.599	-1.447	0.658
	(2.522)	(1.799)	(1.763)	(2.929)	(2.110)	(1.768)
Sharpened q-values	{1.000}	{1.000}	{1.000}	{1.000}		
Observations	837	837	837	837	837	837
Mean dependent variable (Baseline)	24.050	13.628	7.495	24.050	13.628	7.495
Covariates included	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: Maharashtra						
Treatment: Loan	-1.668	1.238	1.236	-7.115	-2.977	-0.381
	(9.320)	(6.416)	(5.531)	(8.924)	(5.789)	(5.449)
Sharpened q-values	{1.000}	{1.000}	{1.000}	{1.000}		
Observations	592	592	592	592	592	592
Mean dependent variable (Baseline)	33.027	23.274	25.314	28.108	20.035	23.380
Covariates included	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table repeats Table D.7. Panel A restricts the sample to Odisha and Panel B to Maharashtra. Sharpened q-values (curly braces) are computed separately within each state panel, so a family consists of the outcomes within one state panel. Standard errors are clustered at the village level. All other details follow Table D.7. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$.

Table D.20: ITT by state: Loan intervention and agricultural expansion

	Real income from farming	Harvest	Yield (Harvest/ area)
	(1)	(2)	(3)
Panel A: Odisha			
Treatment: Loan	148.770	8.666	-2.181
	(154.673)	(6.716)	(5.814)
Sharpened q-values	{1.000}	{1.000}	{1.000}
Observations	701	837	769
Mean dependent variable (Baseline)	0.000	42.032	21.021
Covariates included	Yes	Yes	Yes
Block FE	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes
Panel B: Maharashtra			
Treatment: Loan	941.702	28.082	3.824
	(1012.291)	(27.634)	(3.620)
Sharpened q-values	{0.558}	{0.558}	{0.558}
Observations	564	592	592
Mean dependent variable (Baseline)	2410.705	26.080	5.786
Covariates included	Yes	Yes	Yes
Block FE	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes

Notes: The table repeats Table D.8. Panel A restricts the sample to Odisha and Panel B to Maharashtra. Sharpened q-values (curly braces) are computed separately within each state panel, so a family consists of the outcomes within one state panel. Standard errors are clustered at the village level. All other details follow Table D.8. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$.

Table D.21: ITT by state: Loan intervention and wages

	Aggregate real wages		Total real expenditure (main crops)	
	Male	Female	Male	Female
	(1)	(2)	(3)	(4)
Panel A:Odisha				
Treatment: Loan	-0.103	-0.156	5.739	1.539
	(0.155)	(0.118)	(9.264)	(6.136)
Sharpened q-values	{1.000}	{1.000}	{1.000}	{1.000}
Observations	597	597	837	837
Mean dependent variable (Baseline)	2.474	1.750	49.694	16.091
Covariates included	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes
Panel B: Maharashtra				
Treatment: Loan	0.039	0.073	-22.527	-8.541
	(0.039)	(0.062)	(28.748)	(23.565)
Sharpened q-values	{1.000}	{1.000}	{1.000}	{1.000}
Observations	564	563	592	592
Mean dependent variable (Baseline)	3.148	2.268	59.745	95.166
Covariates included	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes
Survey month FE	Yes	Yes	Yes	Yes

Notes: The table repeats Table D.9. Panel A restricts the sample to Odisha and Panel B to Maharashtra. Sharpened q-values (curly braces) are computed separately within each state panel, so a family consists of the outcomes within one state panel. Standard errors are clustered at the village level. All other details follow Table D.9. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$.

Table D.22: LATE and ITT by state, women: Decision making

	Share of decisions made	Farming	Livestock & Poultry	Non agriculture	Household purchases	Private consumption	Domestic work	Personal activities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Odisha								
LATE: Received	0.319***	0.268**	0.190	0.329***	0.468***	0.438***	0.489***	0.036
loan product (γ_F)	(0.079)	(0.115)	(0.141)	(0.116)	(0.164)	(0.142)	(0.162)	(0.128)
	{0.002}	{0.013}	{0.056}	{0.011}	{0.011}	{0.011}	{0.011}	{0.242}
ITT: Village	0.144***	0.121**	0.086	0.149**	0.212***	0.198***	0.221***	0.016
assigned to	(0.038)	(0.051)	(0.065)	(0.060)	(0.066)	(0.055)	(0.064)	(0.058)
treatment (β_F)	{0.004}	{0.015}	{0.059}	{0.015}	{0.004}	{0.004}	{0.004}	{0.242}
First-stage F	55.088	55.088	55.088	55.088	55.088	55.088	55.088	55.088
Observations	736	736	736	736	736	736	736	736
Control group mean	0.248	0.671	0.267	0.148	0.318	0.284	0.290	0.635
Panel B: Maharashtra								
LATE: Received	-0.342***	-0.894***	-0.156*	-0.836***	-0.178	-0.221	-0.200	-0.276*
loan product (γ_F)	(0.060)	(0.197)	(0.080)	(0.134)	(0.130)	(0.148)	(0.151)	(0.154)
	{0.001}	{0.001}	{0.081}	{0.001}	{0.136}	{0.136}	{0.136}	{0.088}
ITT: Village	-0.194***	-0.506***	-0.088*	-0.473***	-0.101	-0.125	-0.113	-0.156*
assigned to	(0.035)	(0.090)	(0.044)	(0.061)	(0.080)	(0.089)	(0.089)	(0.090)
treatment (β_F)	{0.001}	{0.001}	{0.071}	{0.001}	{0.122}	{0.122}	{0.122}	{0.100}
First-stage F	66.142	66.142	66.142	66.142	66.142	66.142	66.142	66.142
Observations	390	390	390	390	390	390	390	390
Control group mean	0.429	0.701	0.109	0.552	0.688	0.792	0.751	0.900
Baseline controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Interview month fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE and ITT estimates of the loan product on women's involvement in decision making. LATE estimates are from a 2SLS model using village assignment to treatment as an instrument for a household's receipt of the loan product. ITT estimates are from ordinary least squares regressions of the outcome on village assignment to treatment, estimated on the same sample as the corresponding LATE. The dependent variables are listed as column titles. See Table B.1 for description of column titles. Panel A restricts the sample to women in Odisha and Panel B to women in Maharashtra. All regressions control for the household's demographic and economic characteristics from the baseline survey and include block and interview month fixed effects. The first-stage F is the F statistic for regressing village assignment on household's receipt of the loan product, conditional on the controls and fixed effects. Standard errors in all models are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q-values are reported in curly braces. For both corrections, a family consists of the eight outcomes within a panel, and LATE and ITT estimates are treated as separate families. Stars refer to conventional p-values: $*p < 0.1, **p < .05, ***p < .01$

Table D.23: LATE: Loan intervention and joint decision making with spouse

	Farming	Livestock & Poultry	Non agriculture	Household purchases
	(1)	(2)	(3)	(4)
Panel A: Women				
Received loan	-0.081	0.139*	-0.151	0.203*
product	(0.127)	(0.073)	(0.104)	(0.110)
Constant	-0.628***	-0.133	-0.411**	-0.500***
	(0.225)	(0.116)	(0.203)	(0.193)
Sharpened q-values	0.248	0.153	0.153	0.153
Observations	1126	1126	1126	1126
Control group mean	0.350	0.088	0.233	0.295
Panel B: Men				
Received loan	0.042	0.095	0.021	-0.027
product	(0.105)	(0.074)	(0.094)	(0.096)
Constant	-0.375***	-0.288***	-0.149*	-0.003
	(0.136)	(0.109)	(0.090)	(0.115)
Sharpened q-values	1	1	1	1
Observations	1364	1364	1364	1364
Control group mean	0.350	0.088	0.233	0.295
Difference ($\gamma_F - \gamma_M$)	-0.124	0.045	-0.172	0.230
p-value of test ($\gamma_F = \gamma_M$)	0.452	0.669	0.221	0.116
Baseline Controls				
Block fixed effects	Yes	Yes	Yes	Yes
Panel C: Concordance between spouses on joint decision making				
Received loan product	-0.134	-0.043	-0.007	0.061
	(0.092)	(0.061)	(0.044)	(0.054)
Constant	0.088	-0.100	-0.063	0.167
	(0.132)	(0.072)	(0.059)	(0.118)
Sharpened q-values	1	1	1	1
Observations	499	499	499	499
Control group mean	0.106	0.048	0.031	0.035
Baseline Controls	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of loan treatment on decisions made jointly with their spouse. The analysis uses treatment assignment as an exogenous instrument for a household's receipt of the loan product. The outcome variables are binary and take the value 0 if the woman is not involved in the decision or makes the decision on her own. Column 1 includes farming and processing of the harvest for food consumption and high value crop farming. Column 2 includes management of large livestock (e.g. cattle and buffaloes), small livestock (sheep, goats, pigs) and poultry. Column 3 includes non-farm economic activities such as small business, self-employment and wage and salary employment. Column 4 includes decisions on purchase of large items such as bicycles, land and transport vehicles as well as routine items such as food. The sample is restricted to women in the endline survey in Panel A and men in Panel B. Therefore, only one observation per household is analyzed. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.24: LATE: Heterogeneous effects of ex-ante asset ownership on bank account and debit card use

	Own bank account	Deposited money	Withdrew money	Received money	Use debit card alone	Share debit card
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan *	0.037	0.023	0.029	0.030	0.035	-0.038
Asset quintile	(0.050)	(0.055)	(0.048)	(0.050)	(0.059)	(0.054)
Received Loan	-0.139	0.326	0.351*	0.149	0.594**	-0.008
	(0.217)	(0.237)	(0.202)	(0.213)	(0.259)	(0.219)
Asset quintile	-0.020	-0.016	-0.021	-0.014	-0.036*	0.012
	(0.018)	(0.019)	(0.017)	(0.017)	(0.019)	(0.019)
Constant	0.483**	-0.088	-0.149	-0.148	0.694***	0.201
	(0.236)	(0.192)	(0.194)	(0.191)	(0.208)	(0.179)
Sharpened q-values	1.000	1.000	1.000	1.000	1.000	1.000
Observations	1126	1126	1126	1126	1126	1126
Control group mean	0.857	0.374	0.421	0.362	0.178	0.309
Panel B: Men						
Received loan *	0.041	0.115**	0.113**	0.068	-0.075	0.064
Asset quintile	(0.045)	(0.049)	(0.050)	(0.051)	(0.058)	(0.052)
Received Loan	0.020	0.010	-0.058	0.076	0.468**	-0.153
	(0.165)	(0.190)	(0.183)	(0.203)	(0.190)	(0.177)
Asset quintile	0.002	-0.039**	-0.041**	-0.025	0.013	-0.009
	(0.015)	(0.016)	(0.016)	(0.016)	(0.019)	(0.017)
Constant	0.701***	0.283	0.334	0.302	1.178***	-0.003
	(0.177)	(0.315)	(0.250)	(0.246)	(0.178)	(0.139)
Sharpened q-values	0.321	0.074	0.074	0.213	0.213	0.213
Observations	1364	1364	1364	1364	1364	1364
Control group mean	0.836	0.303	0.400	0.217	0.520	0.227

Notes: The table reports heterogeneous LATE estimates by baseline asset quintile, estimated by 2SLS. Loan receipt and its interaction with the baseline asset quintile are treated as endogenous and instrumented with village assignment to treatment and its interaction with the baseline asset quintile. Dependent variables are binary and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.25: LATE: Heterogeneous effects of ex-ante asset ownership on mobile phone and digital finance

	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)
Panel A: Women			
Received loan *	0.027	0.027	0.031
Asset quintile	(0.072)	(0.054)	(0.053)
Received Loan	0.053	0.802***	0.709***
	(0.287)	(0.243)	(0.224)
Asset quintile	-0.014	-0.038**	-0.030*
	(0.027)	(0.017)	(0.016)
Constant	0.723***	0.628***	0.716***
	(0.221)	(0.199)	(0.192)
Sharpened q-values	1.000	1.000	1.000
Observations	794	1126	1126
Control group mean	0.300	0.219	0.184
Panel B: Men			
Received loan *	-0.074	0.016	0.046
Asset quintile	(0.056)	(0.052)	(0.051)
Received Loan	0.294	0.386**	0.240
	(0.186)	(0.171)	(0.173)
Asset quintile	-0.001	-0.005	-0.012
	(0.022)	(0.019)	(0.017)
Constant	0.857***	1.330***	1.449***
	(0.205)	(0.145)	(0.147)
Sharpened q-values	1.000	1.000	1.000
Observations	1007	1364	1364
Control group mean	0.717	0.609	0.582

Notes: The table reports heterogeneous LATE estimates by baseline asset quintile, estimated by 2SLS. Loan receipt and its interaction with the baseline asset quintile are treated as endogenous and instrumented with village assignment to treatment and its interaction with the baseline asset quintile. Dependent variables are binary and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.26: LATE: Heterogeneous effects of ex-ante asset ownership on time use (peak time in agricultural season)

	Farming	Livestock & Poultry	Non-agriculture	Household Purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan *	-0.254	-0.044	-0.273	-0.241**	-0.523**	-0.016
Asset quintile	(0.424)	(0.074)	(0.207)	(0.113)	(0.231)	(0.170)
Received Loan	-1.455	-0.042	-1.262*	0.715	1.314	0.990
	(2.137)	(0.356)	(0.649)	(0.528)	(0.980)	(0.786)
Asset quintile	0.077	0.016	0.166*	0.058*	0.083	0.059
	(0.113)	(0.034)	(0.089)	(0.034)	(0.057)	(0.046)
Sharpened q-values	0.602	0.602	0.335	0.107	0.107	0.857
Observations	1119	1119	1119	1119	453	443
Control group mean	3.130	0.356	1.323	0.677	3.188	1.676
Panel B: Men						
Received loan *	-0.342	-0.254*	0.289	-0.322**	0.389	0.285**
Asset quintile	(0.442)	(0.139)	(0.349)	(0.129)	(0.293)	(0.123)
Received Loan	1.475	0.461	-0.757	0.762**	-0.113	-0.517
	(1.298)	(0.374)	(1.291)	(0.319)	(0.910)	(0.318)
Asset quintile	-0.083	0.071	-0.121	0.006	-0.060	-0.148***
	(0.162)	(0.065)	(0.104)	(0.043)	(0.102)	(0.045)
Sharpened q-values	0.282	0.100	0.282	0.065	0.160	0.065
Observations	1353	1353	1353	1353	311	528
Control group mean	4.144	0.489	1.247	0.677	1.611	1.602

Notes: The table reports heterogeneous LATE estimates by baseline asset quintile, estimated by 2SLS. Loan take-up and its interaction with the baseline asset quintile are treated as endogenous and instrumented with village assignment to treatment and its interaction with the baseline asset quintile. Dependent variables are the number of hours spent in an activity and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.27: LATE: Heterogeneous effects of ex-ante asset ownership on time use (yesterday)

	Farming	Livestock & Poultry	Non-agriculture	Household Purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan *	0.156	-0.083	0.057	-0.544*	0.023	0.025
Asset quintile	(0.351)	(0.092)	(0.233)	(0.291)	(0.259)	(0.176)
Received Loan	-2.647	0.037	-1.136*	1.385	-1.017	0.782
	(1.690)	(0.449)	(0.675)	(1.110)	(1.058)	(0.756)
Asset quintile	-0.061	0.019	0.014	0.131*	0.061	0.048
	(0.094)	(0.039)	(0.084)	(0.075)	(0.074)	(0.044)
Sharpened q-values	1.000	1.000	1.000	0.589	1.000	1.000
Observations	1081	1081	1081	1081	416	408
Control group mean	2.244	0.402	0.953	1.326	3.528	1.788
Panel B: Men						
Received loan *	-0.277	-0.324**	-0.007	-0.572	0.394	0.406**
Asset quintile	(0.377)	(0.156)	(0.297)	(0.362)	(0.261)	(0.190)
Received Loan	1.190	0.602	0.480	1.589	-1.173	-1.116**
	(1.129)	(0.398)	(1.050)	(1.077)	(0.798)	(0.563)
Asset quintile	-0.009	0.107	-0.069	-0.107	-0.097	-0.195***
	(0.126)	(0.077)	(0.092)	(0.080)	(0.099)	(0.075)
Sharpened q-values	0.246	0.130	0.487	0.152	0.152	0.130
Observations	1345	1345	1345	1345	321	536
Control group mean	2.842	0.521	1.272	1.443	1.866	2.016

Notes: The table reports heterogeneous LATE estimates by baseline asset quintile, estimated by 2SLS. Loan receipt and its interaction with the baseline asset quintile are treated as endogenous and instrumented with village assignment to treatment and its interaction with the baseline asset quintile. Dependent variables are the number of hours spent in an activity and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.28: LATE: Heterogeneous effects of ex-ante asset ownership on involvement in decision-making

	Share of decisions made	Farming	Livestock & Poultry	Non agriculture	Consumption	Private consumption	Domestic work	Personal activities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Women								
Received loan *	0.049	0.078	0.036	0.069	0.029	0.014	0.068	0.076
Asset quintile	(0.039)	(0.071)	(0.032)	(0.063)	(0.068)	(0.070)	(0.070)	(0.056)
Received Loan	-0.108	-0.463*	-0.049	-0.365*	0.079	0.108	-0.023	-0.346
	(0.126)	(0.258)	(0.137)	(0.213)	(0.258)	(0.286)	(0.272)	(0.219)
Asset quintile	-0.012	-0.031	-0.008	-0.008	-0.011	-0.012	-0.033*	-0.028
	(0.012)	(0.023)	(0.015)	(0.021)	(0.021)	(0.022)	(0.020)	(0.019)
Constant	-0.024	0.206	0.010	-0.311	-0.041	0.088	0.172	1.079***
	(0.096)	(0.201)	(0.121)	(0.211)	(0.199)	(0.189)	(0.211)	(0.175)
Sharpened q-values	0.790	0.790	0.790	0.790	0.790	0.790	0.790	0.790
Observations	1126	1126	1126	1126	1126	1126	1126	1126
Control group mean	0.317	0.683	0.207	0.302	0.459	0.478	0.466	0.736
Panel B: Men								
Received loan *	-0.023	-0.067	-0.020	-0.003	-0.149**	-0.073	-0.085	-0.049
Asset quintile	(0.024)	(0.051)	(0.055)	(0.065)	(0.061)	(0.060)	(0.065)	(0.046)
Received Loan	0.090	0.142	-0.055	-0.024	0.442**	0.214	0.338	0.281
	(0.072)	(0.137)	(0.154)	(0.210)	(0.203)	(0.227)	(0.215)	(0.182)
Asset quintile	-0.008	-0.018	-0.003	-0.013	-0.018	0.006	-0.003	-0.001
	(0.008)	(0.018)	(0.020)	(0.022)	(0.018)	(0.017)	(0.019)	(0.014)
Constant	0.106	0.539***	-0.092	0.214	0.588***	0.178	0.552***	0.068
	(0.081)	(0.161)	(0.156)	(0.227)	(0.192)	(0.188)	(0.160)	(0.208)
Sharpened q-values	0.658	0.636	0.799	0.830	0.137	0.636	0.636	0.658
Observations	1364	1364	1364	1364	1364	1364	1364	1364
Control group mean	0.273	0.839	0.283	0.322	0.477	0.293	0.434	0.214
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports heterogeneous LATE estimates by baseline asset quintile, estimated by 2SLS. Loan receipt and its interaction with the baseline asset quintile are treated as endogenous and instrumented with village assignment to treatment and its interaction with the baseline asset quintile. Dependent variables are binary and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.29: LATE: Heterogeneous effects of caste on individual bank account and debit card use

	Own bank account	Deposited money	Withdrew money	Received money	Use debit card alone	Share debit card
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan	-0.024	0.243	0.130	0.249	0.169	-0.045
* SC/ST	(0.141)	(0.156)	(0.149)	(0.156)	(0.170)	(0.166)
Received Loan	-0.013	0.321***	0.400***	0.163	0.654***	-0.118
	(0.114)	(0.104)	(0.103)	(0.099)	(0.135)	(0.104)
SC/ST	0.080	-0.034	-0.038	-0.025	-0.043	0.081
	(0.050)	(0.048)	(0.052)	(0.047)	(0.051)	(0.053)
Constant	0.462**	-0.068	-0.144	-0.129	0.636***	0.269
	(0.217)	(0.174)	(0.182)	(0.163)	(0.164)	(0.168)
Sharpened q-values	1.000	0.563	0.624	0.563	0.624	1.000
Observations	1126	1126	1126	1126	1126	1126
Control group mean	0.857	0.374	0.421	0.362	0.178	0.309
Panel B: Men						
Received loan	-0.051	-0.063	-0.052	0.014	-0.024	0.169
* SC/ST	(0.127)	(0.190)	(0.203)	(0.192)	(0.160)	(0.148)
Received Loan	0.155**	0.364***	0.287**	0.270**	0.256**	-0.014
	(0.078)	(0.117)	(0.114)	(0.125)	(0.110)	(0.101)
SC/ST	0.006	0.044	0.057	-0.005	-0.015	-0.028
	(0.035)	(0.058)	(0.062)	(0.059)	(0.052)	(0.046)
Constant	0.645***	0.183	0.253	0.296	1.289***	-0.109
	(0.152)	(0.263)	(0.195)	(0.192)	(0.182)	(0.132)
Sharpened q-values	1.000	1.000	1.000	1.000	1.000	1.000
Observations	1364	1364	1364	1364	1364	1364
Control group mean	0.836	0.303	0.400	0.217	0.520	0.227

Notes: The table reports heterogeneous LATE estimates by household's caste status, estimated by 2SLS. Loan receipt and its interaction with whether the household belonged to Scheduled Caste/ Scheduled Tribe are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same covariate. Dependent variables are binary and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.30: LATE: Heterogeneous effects of caste on mobile phone and digital financial services

	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)
Panel A: Women			
Received loan * SC/ST	-0.057	0.300*	0.360**
	(0.189)	(0.161)	(0.181)
Received Loan	0.157	0.792***	0.693***
	(0.133)	(0.136)	(0.133)
SC/ST	0.016	-0.049	-0.055
	(0.071)	(0.049)	(0.042)
Constant	0.743***	0.585***	0.672***
	(0.194)	(0.173)	(0.161)
Sharpened q-values	0.342	0.103	0.103
Observations	794	1126	1126
Control group mean	0.300	0.219	0.184
Panel B: Men			
Received loan * SC/ST	-0.142	0.133	0.160
	(0.155)	(0.150)	(0.164)
Received Loan	0.125	0.396***	0.330***
	(0.109)	(0.082)	(0.086)
SC/ST	0.091	0.004	-0.002
	(0.056)	(0.045)	(0.049)
Constant	0.963***	1.319***	1.389***
	(0.158)	(0.139)	(0.135)
Sharpened q-values	0.603	0.603	0.603
Observations	1007	1364	1364
Control group mean	0.717	0.609	0.582

Notes: The table reports heterogeneous LATE estimates by household's caste status, estimated by 2SLS. Loan receipt and its interaction with whether the household belonged to Scheduled Caste/ Scheduled Tribe are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same covariate. Dependent variables are binary and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.31: LATE: Heterogeneous effects of caste on time use (peak time in agricultural season)

	Farming	Livestock & Poultry	Non-agriculture	Household Purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan *	0.600	-0.400	0.059	0.249	-0.660	0.836
SC/ST	(1.388)	(0.261)	(0.649)	(0.318)	(0.913)	(0.630)
Received Loan	-2.473**	-0.071	-2.189***	-0.162	-0.200	0.627**
	(0.960)	(0.200)	(0.557)	(0.257)	(0.462)	(0.311)
SC/ST	0.173	0.119	0.134	0.059	0.176	-0.187
	(0.362)	(0.093)	(0.245)	(0.111)	(0.234)	(0.128)
Constant	0.677	0.393	-1.235	-0.794*	4.178***	1.580***
	(1.219)	(0.329)	(0.907)	(0.434)	(1.005)	(0.593)
Sharpened q-values	1.000	1.000	1.000	1.000	1.000	1.000
Observations	1119	1119	1119	1119	453	443
Control group mean	3.130	0.356	1.323	0.677	3.188	1.676
Panel B: Men						
Received loan *	0.763	-0.534	0.770	0.516	1.619*	0.608
SC/ST	(1.506)	(0.634)	(0.852)	(0.315)	(0.977)	(0.512)
Received Loan	0.289	-0.139	-0.128	-0.313	0.535	0.101
	(0.783)	(0.194)	(0.507)	(0.240)	(0.444)	(0.200)
SC/ST	-0.532	0.286	-0.343	-0.238**	-0.229	-0.291**
	(0.502)	(0.283)	(0.290)	(0.095)	(0.232)	(0.124)
Constant	1.677	-0.548	2.993***	1.750***	2.165**	1.666***
	(1.228)	(0.430)	(0.898)	(0.374)	(0.868)	(0.493)
Sharpened q-values	0.692	0.471	0.471	0.435	0.435	0.457
Observations	1353	1353	1353	1353	311	528
Control group mean	4.144	0.489	1.247	0.677	1.611	1.602
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports heterogeneous LATE estimates by household's caste status, estimated by 2SLS. Loan receipt and its interaction with whether the household belonged to Scheduled Caste/ Scheduled Tribe are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same covariate. Dependent variables are the number of hours spent in activity and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.32: LATE: Heterogeneous effects of caste time use (yesterday)

	Farming	Livestock & Poultry	Non-agriculture	Household Purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan *	0.862	-0.405	0.317	0.599	-0.058	0.731
SC/ST	(1.121)	(0.316)	(0.670)	(0.720)	(0.801)	(0.515)
Received Loan	-2.422***	-0.107	-1.058*	-0.567	-0.924*	0.578
	(0.791)	(0.196)	(0.592)	(0.596)	(0.516)	(0.355)
SC/ST	-0.102	0.188*	0.098	0.212	-0.149	-0.226
	(0.292)	(0.105)	(0.196)	(0.252)	(0.263)	(0.148)
Constant	-0.363	0.060	-1.332*	-0.275	4.679***	2.060***
	(0.880)	(0.312)	(0.766)	(1.172)	(0.989)	(0.599)
Sharpened q-values	1.000	1.000	1.000	1.000		
Observations	1081	1081	1081	1081	416	408
Control group mean	2.244	0.402	0.953	1.326	3.528	1.788
Panel B: Men						
Received loan *	-0.444	-0.574	0.788	1.084	1.755**	0.850
SC/ST	(1.184)	(0.688)	(0.651)	(0.766)	(0.894)	(0.616)
Received Loan	0.510	-0.190	0.250	-0.371	-0.560	-0.209
	(0.711)	(0.192)	(0.430)	(0.675)	(0.465)	(0.235)
SC/ST	-0.359	0.288	-0.360*	-0.403**	-0.270	-0.259
	(0.400)	(0.301)	(0.214)	(0.201)	(0.221)	(0.193)
Constant	1.339	-0.601	2.051**	5.632***	2.039***	2.732***
	(0.860)	(0.518)	(0.847)	(1.194)	(0.707)	(0.676)
Sharpened q-values	0.513	0.502	0.429	0.429		
Observations	1345	1345	1345	1345	321	536
Control group mean	2.842	0.521	1.272	1.443	1.866	2.016
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports heterogeneous LATE estimates by household's caste status, estimated by 2SLS. Loan receipt and its interaction with whether the household belonged to Scheduled Caste/ Scheduled Tribe are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same covariate. Dependent variables are the number of hours spent in activity and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.33: LATE: Heterogeneous effects of caste on participation in decisions

	Share of decisions made	Farming	Livestock & Poultry	Non agriculture	Consumption	Private consumption	Domestic work	Personal activities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Women								
Received loan *	0.124	0.311*	-0.064	0.140	0.269	0.230	0.208	0.092
SC/ST	(0.109)	(0.183)	(0.156)	(0.173)	(0.194)	(0.190)	(0.202)	(0.171)
Received Loan	0.007	-0.315**	0.087	-0.190	0.082	0.079	0.129	-0.132
	(0.080)	(0.135)	(0.091)	(0.132)	(0.134)	(0.124)	(0.142)	(0.121)
SC/ST	0.039	0.005	0.065*	0.051	0.077	0.084	0.046	0.022
	(0.032)	(0.069)	(0.039)	(0.052)	(0.061)	(0.068)	(0.062)	(0.057)
Constant	-0.049	0.150	0.002	-0.348	-0.057	0.073	0.120	1.016***
	(0.114)	(0.204)	(0.122)	(0.229)	(0.200)	(0.199)	(0.215)	(0.160)
Sharpened q-values	0.941	0.941	0.941	0.941				
Observations	1126	1126	1126	1126	1126	1126	1126	1126
Control group mean	0.317	0.683	0.207	0.302	0.459	0.478	0.466	0.736
Panel B: Men								
Received loan *	-0.022	0.019	0.054	-0.037	0.028	-0.188	-0.223	-0.045
SC/ST	(0.101)	(0.175)	(0.164)	(0.184)	(0.204)	(0.230)	(0.215)	(0.174)
Received Loan	0.028	-0.059	-0.130	-0.022	-0.002	0.052	0.150	0.150*
	(0.054)	(0.081)	(0.096)	(0.110)	(0.128)	(0.124)	(0.133)	(0.085)
SC/ST	0.007	-0.042	-0.016	-0.023	-0.047	0.113	0.069	0.008
	(0.026)	(0.068)	(0.056)	(0.053)	(0.060)	(0.073)	(0.070)	(0.056)
Constant	0.147*	0.625***	-0.051	0.227	0.757***	0.291*	0.669***	0.160
	(0.076)	(0.155)	(0.130)	(0.207)	(0.195)	(0.163)	(0.145)	(0.171)
Sharpened q-values	1.000	1.000	1.000	1.000				
Observations	1364	1364	1364	1364	1364	1364	1364	1364
Control group mean	0.273	0.839	0.283	0.322	0.477	0.293	0.434	0.214
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports heterogeneous LATE estimates by household's caste status, estimated by 2SLS. Loan receipt and its interaction with whether the household belonged to Scheduled Caste/ Scheduled Tribe are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same covariate. Dependent variables are binary and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.34: LATE: Heterogeneous effects of ex-ante spousal coordination on involvement in decisions

	Share of decisions made	Farming	Livestock & Poultry	Non agriculture	Consumption	Private consumption	Domestic work	Personal activities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Women								
Received loan * BL	0.629***	0.632	0.404	0.808***	1.052**	0.904**	0.996*	0.611**
share of joint decisions	(0.230)	(0.390)	(0.267)	(0.258)	(0.497)	(0.453)	(0.535)	(0.276)
Received Loan	0.000	-0.260**	0.042	-0.214*	0.087	0.089	0.126	-0.156
	(0.070)	(0.125)	(0.101)	(0.124)	(0.123)	(0.115)	(0.123)	(0.114)
Baseline Share	-0.134**	-0.156	-0.162	-0.057	-0.167	-0.230	-0.267*	-0.027
of joint decisions	(0.066)	(0.165)	(0.123)	(0.106)	(0.181)	(0.146)	(0.145)	(0.126)
Constant	-0.065	0.144	-0.019	-0.366*	-0.079	0.051	0.093	1.001***
	(0.105)	(0.180)	(0.123)	(0.210)	(0.189)	(0.191)	(0.207)	(0.156)
Sharpened q-values	0.022	0.073	0.073	0.017	0.054	0.059	0.068	0.054
Observations	1126	1126	1126	1126	1126	1126	1126	1126
Control group mean	0.317	0.683	0.207	0.302	0.459	0.478	0.466	0.736
Min. eigenvalue statistic								
Panel B: Men								
Received loan * BL	0.144	-0.275	0.118	1.040***	-0.246	0.316	0.242	-0.383
share of joint decisions	(0.169)	(0.273)	(0.408)	(0.372)	(0.434)	(0.418)	(0.406)	(0.360)
Received Loan	0.017	-0.036	-0.130	-0.089	0.021	-0.010	0.081	0.164
	(0.063)	(0.092)	(0.103)	(0.126)	(0.140)	(0.134)	(0.136)	(0.105)
Baseline Share	-0.115	0.102	0.087	-0.540***	0.097	-0.279	-0.235	0.101
of joint decisions	(0.081)	(0.127)	(0.221)	(0.167)	(0.232)	(0.217)	(0.218)	(0.195)
Constant	0.158**	0.610***	-0.049	0.292	0.744***	0.310*	0.681***	0.139
	(0.076)	(0.163)	(0.131)	(0.186)	(0.205)	(0.167)	(0.139)	(0.171)
Sharpened q-values	1.000	1.000	1.000	0.042	1.000	1.000	1.000	1.000
Observations	1364	1364	1364	1364	1364	1364	1364	1364
Control group mean	0.273	0.839	0.283	0.322	0.477	0.293	0.434	0.214
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports heterogeneous LATE estimates by baseline spousal coordination, estimated by 2SLS. Loan receipt and its interaction with the share of decisions farmer reported making jointly with spouse in the baseline survey are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same covariate. Dependent variables are binary and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.35: LATE: Heterogeneous effects of ex-ante spousal coordination on time use (peak time in agricultural season)

	Farming	Livestock & Poultry	Non-agriculture	Household Purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan * BL	0.561	1.244**	1.660*	0.973	4.311**	2.074
share of joint decisions	(2.214)	(0.574)	(0.922)	(1.162)	(2.073)	(1.926)
Received Loan	-2.149**	-0.279	-1.033*	-0.425	-1.039**	0.862**
	(0.868)	(0.212)	(0.554)	(0.563)	(0.513)	(0.366)
Baseline Share	-0.540	-0.491**	-0.396	-0.186	-0.317	-0.841*
of joint decisions	(0.969)	(0.245)	(0.345)	(0.322)	(0.830)	(0.473)
Constant	-0.329	0.002	-1.350*	-0.258	4.584***	2.243***
	(0.850)	(0.292)	(0.754)	(1.224)	(1.040)	(0.653)
Sharpened q-values	0.667	0.128	0.128	0.319	0.128	0.268
Observations	1081	1081	1081	1081	416	408
Control group mean	2.244	0.402	0.953	1.326	3.528	1.788
Panel B: Men						
Received loan * BL	-2.773	0.357	1.073	1.179	-4.112	1.637
share of joint decisions	(2.035)	(0.978)	(1.491)	(1.713)	(2.510)	(1.206)
Received Loan	0.529	-0.365	0.406	-0.145	0.165	-0.074
	(0.699)	(0.289)	(0.445)	(0.633)	(0.411)	(0.236)
Baseline Share	1.772*	0.101	-0.622	-0.362	1.842	-1.498**
of joint decisions	(1.070)	(0.528)	(0.737)	(0.825)	(1.421)	(0.615)
Constant	1.146	-0.620	2.146**	5.726***	2.616***	3.043***
	(0.854)	(0.507)	(0.877)	(1.280)	(0.785)	(0.676)
Sharpened q-values	0.537	0.557	0.537	0.537	0.537	0.537
Observations	1345	1345	1345	1345	321	536
Control group mean	2.842	0.521	1.272	1.443	1.866	2.016

Notes: The table reports heterogeneous LATE estimates by baseline spousal coordination, estimated by 2SLS. Loan receipt and its interaction with the share of decisions farmer reported making jointly with spouse in the baseline survey are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same covariate. Dependent variables are hours spent in activity listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.36: LATE: Heterogeneous effects of ex-ante spousal coordination on time use (yesterday)

	Farming	Livestock & Poultry	Non-agriculture	Household Purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan * BL	1.799	0.603	1.998**	1.572**	7.582***	-0.166
share of joint decisions	(2.859)	(0.543)	(0.920)	(0.663)	(2.655)	(1.801)
Received Loan	-2.317**	-0.215	-2.267***	-0.150	-0.527	0.944**
	(1.063)	(0.180)	(0.568)	(0.252)	(0.429)	(0.393)
Baseline Share	-1.190	-0.243	-0.267	-0.375**	-1.659**	-0.109
of joint decisions	(1.171)	(0.251)	(0.295)	(0.175)	(0.775)	(0.489)
Constant	0.664	0.349	-1.266	-0.809*	3.760***	1.765***
	(1.160)	(0.305)	(0.907)	(0.437)	(0.944)	(0.603)
Sharpened q-values	0.466	0.251	0.053	0.047	0.027	0.668
Observations	1119	1119	1119	1119	453	443
Control group mean	3.130	0.356	1.323	0.677	3.188	1.676
Panel B: Men						
Received loan * BL	-1.674	0.007	1.861	1.247*	-4.702**	1.305
share of joint decisions	(1.851)	(0.905)	(1.391)	(0.703)	(2.369)	(1.493)
Received Loan	0.581	-0.287	-0.022	-0.242	1.305**	0.201
	(0.857)	(0.243)	(0.511)	(0.225)	(0.518)	(0.195)
Baseline Share	0.941	0.286	-0.601	-0.418	1.902	-0.964
of joint decisions	(0.881)	(0.503)	(0.636)	(0.358)	(1.192)	(0.686)
Constant	1.587	-0.573	3.105***	1.826***	2.626***	1.838***
	(1.250)	(0.426)	(0.932)	(0.397)	(0.826)	(0.486)
Sharpened q-values	0.441	0.847	0.319	0.297	0.297	0.441
Observations	1353	1353	1353	1353	311	528
Control group mean	4.144	0.489	1.247	0.677	1.611	1.602

Notes: The table reports heterogeneous LATE estimates by baseline spousal coordination, estimated by 2SLS. Loan receipt and its interaction with the share of decisions farmer reported making jointly with spouse in the baseline survey are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same covariate. Dependent variables are hours spent in activity listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.37: LATE: Heterogeneous effects of ex-ante spousal coordination on bank account and debit card use

	Own bank account (1)	Deposited money (2)	Withdrew money (3)	Received money (4)	Use debit card alone (5)	Share debit card (6)
			Panel A: Women			
Received loan	-0.570**	0.025	-0.255	-0.126	-0.463	0.309
* BL share of joint decisions	(0.244)	(0.417)	(0.423)	(0.280)	(0.535)	(0.300)
Received Loan	0.023	0.400***	0.467***	0.243**	0.762***	-0.155
	(0.103)	(0.113)	(0.105)	(0.107)	(0.117)	(0.098)
Baseline Share of joint decisions	0.102	-0.038	-0.022	0.165	-0.136	-0.088
	(0.090)	(0.146)	(0.124)	(0.103)	(0.102)	(0.099)
Constant	0.481**	-0.055	-0.129	-0.106	0.657***	0.254
	(0.214)	(0.174)	(0.188)	(0.169)	(0.175)	(0.171)
Sharpened q-values	0.134	1.000	1.000	1.000	1.000	1.000
Observations	1126	1126	1126	1126	1126	1126
Control group mean	0.857	0.374	0.421	0.362	0.178	0.309
			Panel B: Men			
Received loan	0.204	0.350	0.243	0.022	0.184	-0.447
* BL share of joint decisions	(0.320)	(0.472)	(0.413)	(0.396)	(0.416)	(0.360)
Received Loan	0.140*	0.336**	0.261**	0.270**	0.251**	0.055
	(0.082)	(0.134)	(0.128)	(0.136)	(0.103)	(0.098)
Baseline Share of joint decisions	-0.280*	-0.325	-0.152	0.032	-0.299	0.276
	(0.155)	(0.202)	(0.188)	(0.180)	(0.231)	(0.187)
Constant	0.666***	0.210	0.267	0.295	1.311***	-0.133
	(0.139)	(0.246)	(0.192)	(0.202)	(0.181)	(0.130)
Sharpened q-values	1.000	1.000	1.000	1.000	1.000	1.000
Observations	1364	1364	1364	1364	1364	1364
Control group mean	0.836	0.303	0.400	0.217	0.520	0.227

Notes: The table reports heterogeneous LATE estimates by baseline spousal coordination, estimated by 2SLS. Loan receipt and its interaction with the share of decisions farmer reported making jointly with spouse in the baseline survey are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same covariate. Dependent variables are binary and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.38: LATE: Heterogeneous effects of ex-ante spousal coordination on mobile phone and digital finance

	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)
Panel A: Women			
Received loan * BL	0.089	-0.228	-0.404
share of joint decisions	(0.401)	(0.516)	(0.407)
Received Loan	0.139	0.912***	0.846***
	(0.133)	(0.118)	(0.110)
Baseline Share	-0.152	-0.024	-0.001
of joint decisions	(0.227)	(0.105)	(0.106)
Constant	0.738***	0.610***	0.706***
	(0.196)	(0.181)	(0.183)
Sharpened q-values	1.000	1.000	1.000
Observations	794	1126	1126
Control group mean	0.300	0.219	0.184
	(1)	(2)	(3)
Panel B: Men			
Received loan * BL	-0.498	-0.352	-0.416
share of joint decisions	(0.668)	(0.307)	(0.327)
Received Loan	0.115	0.462***	0.402***
	(0.105)	(0.084)	(0.090)
Baseline Share	0.000	0.015	0.129
of joint decisions	(0.377)	(0.156)	(0.178)
Constant	0.963***	1.311***	1.374***
	(0.159)	(0.141)	(0.140)
Sharpened q-values	0.608	0.608	0.608
Observations	1007	1364	1364
Control group mean	0.717	0.609	0.582

Notes: The table reports heterogeneous LATE estimates by baseline spousal coordination, estimated by 2SLS. Loan receipt and its interaction with the share of decisions farmer reported making jointly with spouse in the baseline survey are treated as endogenous and instrumented with village assignment to treatment and its interaction with the same covariate. Dependent variables are binary and listed as column titles. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.39: Baseline characteristics by loan application status

	(1) All	(2) Treatment - Received loan	(3) Treatment - Denied loan	(4) Treatment - Didn't apply	(5) Control	F-test for balance across all groups	(2)-(5) Mean Difference	(3)-(5) Mean Difference	(4)-(5) Mean Difference
Respondent's age	43.301	43.168	42.523	44.929	42.727	3.108**	0.441	-0.204	2.202**
	(-0.447)	(-0.727)	(-1.817)	(-0.725)	(-0.469)	0.031			
Respondent is female	0.367	0.349	0.326	0.406	0.363	0.514	-0.013	-0.037	0.044
	(-0.042)	(-0.061)	(-0.112)	(-0.052)	(-0.067)	0.674			
Respondent is married	0.934	0.94	0.907	0.952	0.927	1.042	0.014	-0.02	0.025
	(-0.01)	(-0.015)	(-0.034)	(-0.011)	(-0.018)	0.379			
Can read and write	0.875	0.872	0.988	0.839	0.88	13.875***	-0.007	0.109***	-0.041
	(-0.016)	(-0.021)	(-0.011)	(-0.033)	(-0.023)	0			
Completed at least primary school	0.788	0.79	0.895	0.71	0.809	4.791***	-0.019	0.086*	-0.099**
	(-0.022)	(-0.036)	(-0.034)	(-0.039)	(-0.031)	0.004			
Completed at least	0.457	0.432	0.558	0.352	0.505	2.841**	-0.073	0.053	-0.154**
secondary school	(-0.029)	(-0.042)	(-0.089)	(-0.049)	(-0.044)	0.043			
Respondent belongs	0.358	0.335	0.198	0.506	0.323	7.771***	0.012	-0.125	0.183***
to SC/ST category	(-0.035)	(-0.051)	(-0.085)	(-0.051)	(-0.046)	0			
Annual Household	0.474	0.494	0.291	0.59	0.433	3.333**	0.061	-0.142	0.157*
Income(<90,000)	(-0.043)	(-0.067)	(-0.098)	(-0.061)	(-0.063)	0.024			
Annual Household Income	0.237	0.259	0.244	0.206	0.239	0.327	0.019	0.005	-0.033
(90,000 - 120,000)	(-0.023)	(-0.044)	(-0.067)	(-0.038)	(-0.034)	0.806			
Annual Household	0.289	0.247	0.465	0.203	0.327	3.982**	-0.08	0.138	-0.124*
Income(>120,000)	(-0.038)	(-0.058)	(-0.085)	(-0.049)	(-0.053)	0.011			
Owns land	0.479	0.483	0.314	0.555	0.464	2.637*	0.019	-0.15	0.091
	(-0.037)	(-0.065)	(-0.107)	(-0.039)	(-0.054)	0.055			
Asset quintile	2.986	3.108	2.488	3.123	2.924	1.796	0.184	-0.435	0.199
	(-0.114)	(-0.201)	(-0.352)	(-0.13)	(-0.169)	0.155			
Respondent has bank	0.933	0.946	0.988	0.894	0.937	7.711***	0.009	0.052***	-0.043*
account	(-0.01)	(-0.016)	(-0.012)	(-0.019)	(-0.014)	0			
Borrowed money last	0.439	0.474	0.512	0.329	0.463	2.451*	0.012	0.049	-0.134**
12 months	(-0.034)	(-0.062)	(-0.103)	(-0.044)	(-0.05)	0.07			
Has outstanding loan	0.212	0.23	0.244	0.213	0.198	0.089	0.032	0.046	0.015
	(-0.035)	(-0.06)	(-0.102)	(-0.053)	(-0.048)	0.966			
F-test of joint significance (F-stat)							0.683	1.044	2.116**
N	1429	352	86	310	681				

Notes: The table reports averages of respondent and household characteristics from the baseline survey by category of application status. Column 1 reports pooled estimates for the full sample. Column 2 reports the averages for households in the loan treatment arm that applied for the loan and received it. Column 3 describes households in the treatment arm that were unsuccessful in getting the loan and column 4 includes households in the treatment arm that did not apply for the loan. Column 5 describes households in the control arm. The fifth column reports the f test for whether each variable is balanced across arms and the last column provides coefficients of a t-test comparing the treatment and control averages. "SC/ST" is abbreviation for Scheduled Caste/ Scheduled Tribe. Standard errors are clustered at the level of treatment assignment (village) and reported below mean values for each variable. The number of observations are included in the last row. $*$ p< 0.1, $**$ p< .05, $***$ p< .01

Table D.40: LATE: Independent bank account and debit card use re-weighted by propensity to borrow

	Own bank account (1)	Deposited money (2)	Withdrew money (3)	Received money (4)	Use debit card alone (5)	Share debit card (6)
Panel A: Women						
Received loan product	-0.139	0.429***	0.464***	0.238**	0.634***	-0.228*
	(0.141)	(0.122)	(0.123)	(0.119)	(0.109)	(0.125)
Constant	0.570**	-0.234	-0.389*	-0.340*	0.720***	0.242
	(0.273)	(0.207)	(0.199)	(0.180)	(0.170)	(0.212)
Sharpened q-values	0.091	0.001	0.001	0.035	0.001	0.044
Observations	1126	1126	1126	1126	1126	1126
Control group mean	0.857	0.374	0.421	0.362	0.178	0.309
Panel B: Men						
Received loan product	0.070	0.354**	0.298*	0.318**	0.090	0.236**
	(0.095)	(0.175)	(0.158)	(0.161)	(0.110)	(0.107)
Constant	0.618***	-0.054	0.004	0.059	1.682***	-0.310*
	(0.167)	(0.309)	(0.247)	(0.222)	(0.202)	(0.162)
Sharpened q-values	0.182	0.098	0.098	0.098	0.182	0.098
Observations	1364	1364	1364	1364	1364	1364
Control group mean	0.836	0.303	0.400	0.217	0.520	0.227
Baseline Controls	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on bank account ownership and use of debit card. Each household is weighted by the inverse of their ex-ante propensity to borrow. The analysis uses treatment assignment as an exogenous instrument for a household's receipt of the loan product. The dependent variables (binary) are listed as column titles. Columns 2 and 3 are conditional on respondent owning an individual bank account. Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.41: LATE: Mobile phone and digital financial services re-weighted by propensity to borrow

	Smartphone	Knows mobile app banking	Uses mobile app banking
	(1)	(2)	(3)
Panel A: Women			
Received loan product	0.047	0.758***	0.744***
	(0.129)	(0.127)	(0.119)
Sharpened q-values	0.315	0.001	0.001
Observations	794	1126	1126
Control group mean	0.300	0.219	0.184
Panel B: Men			
Received loan product	0.005	0.448***	0.396***
	(0.113)	(0.103)	(0.104)
Constant	0.951***	1.526***	1.580***
	(0.201)	(0.154)	(0.156)
Sharpened q-values	0.473	0.001	0.001
Observations	1007	1364	1364
Control group mean	0.717	0.609	0.582
Baseline Controls	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on mobile phone ownership and uptake of digital financial services. Each household is weighted by the inverse of their ex-ante propensity to borrow. The analysis uses treatment assignment as an exogenous instrument for a household's receipt of the loan product. The dependent variables (binary) are listed as column titles. The analysis is conditional on respondent owning an individual bank account. Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.42: LATE: Time use (peak agricultural season) re-weighted by propensity to borrow

	Farming	Livestock & Poultry	Non-agriculture	Household Purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan product	-2.060*	0.023	-2.279***	-0.087	-0.756	1.138***
	(1.069)	(0.240)	(0.637)	(0.254)	(0.589)	(0.372)
Constant	1.467	0.898*	-1.824	-0.942	4.465***	0.623
	(1.280)	(0.506)	(1.196)	(0.603)	(1.166)	(0.691)
Sharpened q-values	0.078	0.445	0.001	0.429	0.177	0.006
Observations	1119	1119	1119	1119	453	443
Control group mean	3.130	0.356	1.323	0.677	3.188	1.676
Panel B: Men						
Received loan product	0.493	-0.763**	-0.323	-0.231	1.949***	0.481**
	(1.053)	(0.375)	(0.652)	(0.212)	(0.691)	(0.217)
Constant	3.164**	-0.571	2.273**	1.655***	2.135*	2.300***
	(1.396)	(0.414)	(0.954)	(0.349)	(1.255)	(0.510)
Sharpened q-values	0.471	0.076	0.471	0.263	0.031	0.073
Observations	1353	1353	1353	1353	311	528
Control group mean	4.144	0.489	1.247	0.677	1.611	1.602
Baseline Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on time use in a busy day of the agricultural season. Each household is weighted by the inverse of their ex-ante propensity to borrow. The analysis uses treatment assignment as an exogenous instrument for a household's receipt of the loan product. The dependent variables (binary) are listed as column titles (see Table B.1 for description of the dependent variables). Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.43: LATE: Time use (yesterday) re-weighted by propensity to borrow

	Farming	Livestock & Poultry	Non-agriculture	Household Purchases	Domestic work	Personal activity
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Women						
Received loan product	-2.000**	-0.066	-0.817	0.168	-1.012*	0.995***
	(0.847)	(0.297)	(0.583)	(0.705)	(0.583)	(0.339)
Constant	-0.809	0.531	-1.588	-0.531	4.058***	1.630**
	(1.056)	(0.559)	(1.138)	(1.192)	(0.989)	(0.677)
Sharpened q-values	0.048	0.379	0.192	0.379	0.125	0.019
Observations	1081	1081	1081	1081	416	408
Control group mean	2.244	0.402	0.953	1.326	3.528	1.788
Panel B: Men						
Received loan product	-0.185	-0.938*	0.342	-0.010	-0.154	0.009
	(0.760)	(0.511)	(0.437)	(0.512)	(0.545)	(0.292)
Constant	1.967**	-0.927	1.249	5.544***	1.633*	3.480***
	(0.876)	(0.676)	(0.802)	(1.154)	(0.920)	(0.694)
Sharpened q-values	1.000	0.673	1.000	1.000	1.000	1.000
Observations	1345	1345	1345	1345	321	536
Control group mean	2.842	0.521	1.272	1.443	1.866	2.016
Baseline Covariates	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on time use on day prior to survey (yesterday). Each household is weighted by the inverse of their ex-ante propensity to borrow. The analysis uses treatment assignment as an exogenous instrument for a household's receipt of the loan product. The dependent variables (binary) are listed as column titles (see Table B.1 for description of the dependent variables). Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses. Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01

Table D.44: LATE: Involvement in decision-making re-weighted by propensity to borrow

	Share of decisions made	Farming	Livestock & Poultry	Non agriculture	Household purchases	Private consumption	Domestic work	Personal activities
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Women								
Received loan product	0.066	-0.245*	0.041	-0.164	0.213	0.141	0.267**	-0.021
	(0.075)	(0.137)	(0.108)	(0.139)	(0.139)	(0.115)	(0.123)	(0.123)
Constant	-0.063	0.263	0.094	-0.420*	-0.042	0.050	0.158	0.954***
	(0.121)	(0.211)	(0.150)	(0.243)	(0.223)	(0.205)	(0.236)	(0.182)
Sharpened q-values	0.513	0.344	0.620	0.413	0.413	0.413	0.303	0.620
Observations	1126	1126	1126	1126	1126	1126	1126	1126
Control group mean	0.317	0.683	0.207	0.302	0.459	0.478	0.466	0.736
Panel B: Men								
Received loan product	-0.028	0.008	-0.140	-0.143	-0.132	-0.134	-0.053	0.047
	(0.069)	(0.129)	(0.128)	(0.144)	(0.137)	(0.131)	(0.121)	(0.145)
Constant	0.008	0.734***	-0.149	-0.020	0.638***	-0.014	0.525***	0.018
	(0.096)	(0.172)	(0.142)	(0.243)	(0.204)	(0.161)	(0.169)	(0.191)
Sharpened q-values	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
Observations	1364	1364	1364	1364	1364	1364	1364	1364
Control group mean	0.273	0.839	0.283	0.322	0.477	0.293	0.434	0.214
Baseline covariates	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Block fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: The table reports LATE estimates of treatment on involvement in production and non-production decisions. Each household is weighted by the inverse of their ex-ante propensity to borrow. The analysis uses treatment assignment as an exogenous instrument for a household's receipt of the loan product. The dependent variables (binary) are listed as column titles (see Table B.1 for description of the dependent variables). Panel A restricts the sample to women in the endline survey and Panel B to men. The estimation controls for the household's demographic and economic characteristics from the baseline survey and includes block fixed effects. Standard errors are clustered at the level of treatment assignment (village) and reported in parentheses.Sharpened q values are computed separately for all outcomes within each gender panel, therefore, a family consists of the outcomes within a gender panel. Stars refer to conventional p-values: $*$ p$<$ 0.1, $**$ p$<$ .05, $***$ p$<$ .01